

Reinforcement learning-based robust formation control for Multi-UAV systems with switching communication topologies

Hongsheng Sha^a, Rongwei Guo^b, Jin Zhou^{c,e}, Xiaojin Zhu^a, Jinchun Ji^d, Zhonghua Miao^{a,*}

^a School of Mechatronic Engineering and Automation, Shanghai University, 200072, Shanghai, China

^b School of Mathematics and Statistics, Qilu University of Technology (Shandong Academy of Sciences), Jinan, 250353, China

^c Shanghai Institute of Applied Mathematics and Mechanics, and Shanghai Key Laboratory of Mechanics in Energy Engineering, Shanghai University, 200072, Shanghai, China

^d Faculty of Engineering and IT, University of Technology Sydney, New South Walls, Australia

^e Shanghai Institute of Aircraft Mechanics and Control, Shanghai, 200092, China

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ABSTRACT

This paper introduces a novel optimal robust formation method for quadcopter multiple unmanned aerial vehicle (multi-UAV) systems. Firstly, a reinforcement learning (RL) algorithm based on a unique gradient descent training approach is proposed to solve the Hamilton–Jacobi–Bellman (HJB) equation, which can effectively eliminate the requirement of the Persistent Excitation (PE) condition. Secondly, the robustness of the controlled system is emphasized, and an Uncertainty and Disturbance Estimator (UDE) observer is developed to suppress model uncertainty and external disturbances through filtering techniques. Furthermore, a switched sliding mode control technique according to the average dwell time (ADT) is employed to convert switching communication topology between UAVs dynamically, and the stability analysis of the corresponding closed-loop control systems is then performed by the use of Lyapunov analysis. Finally, the simulation examples are provided to verify the effectiveness of the designed control strategy.

1. Introduction

With the great advance of information technique and artificial intelligence in recent years, the formation control of multi-UAV systems has recently become a very important yet challenging issue in the fields of coordinated control of autonomous unmanned systems. This is mainly because the UAV possesses some potential advantages of low cost, convenient operation, and security characteristics, and is capable of high agility and rapid maneuvering in complex task environments. Such as detecting crop growth rates [1] and assessing forest fire risks in specific areas [2]. More importantly, the cooperative formation flight of multi-UAV systems is an increasingly developed trend of unmanned aerial applications in military and civil fields, it can accomplish large-scale, high-precision complicated missions that cannot be effectively performed by a single UAV, including fire inspection, emergency search and rescue, and agricultural environment monitoring [3–5]. As a result, more and more strategies have recently been proposed and developed to cope with the formation control problems for multi-UAV systems from various perspectives. Among them, the quadcopter formation control [6], as the most typical one, has developed into key research.

Since the quadrotor is limited in battery power during the formation flight, it is very essential to develop advance optimized control strategy to ensure the endurance of autonomous formation flight for quadrotor multi-UAV systems. From the view of mathematics, a key issue of optimized control in formation flight for quadrotor multi-UAV systems is in essence to establish and solve the Hamilton–Jacobi–Bellman (HJB) optimization equations [7] ((become the Riccati equation [8] for the linear cases) governed by the performance index functions, which are composed of the state vector, control variables, and dynamic model of the controlled system. The goal of optimal control is to determine a precise solution that achieves the objective of rendering the HJB equation equivalent to zero. Actually, quadcopter is a typical underactuated system with high nonlinear and strong coupling [9,10]. Therefore, it is almost impossible to provide accurate solutions through analytical methods, making it difficult to design an effective optimized formation controller.

In recent years, many scholars have attempted to develop various intelligent algorithms for the optimal control problems of quadrotor multi-UAV systems, thereby some effective optimal algorithms have been proposed to deal with the solution of HJB equations, such as data-driven and adaptive

* Corresponding author.

E-mail address: zhonghuamiao@163.com (Z. Miao).

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dynamic programming RL algorithms, as shown in Refs. [11–13]. Both demonstrate good optimization results, but the dependence on PE condition of the input signal increasing the computational burden. In Ref. [14], another RL optimization algorithm was proposed for a class of nonlinear strict-feedback systems, where a critical network is used to approximate the nonlinear term of the HJB equation, and an actor network is utilized to optimize the control input. For this purpose, a specialized gradient descent method is proposed to train the critic-actor loop until the Bellman equation is balanced. This method does not require continuous action of the input signal, thus releasing the PE condition. Subsequently, this simple and feasible optimal strategy is developed into consensus control for a general second-order multi-agent system [15], which inspires our present study for quadrotor multi-UAV systems.

In general, it is quite difficult to obtain an accurate model of quadrotor in practice because of its complex aerodynamic characteristics, thruster saturation, and various random environmental disturbances such as wind gusts, or load changes. Therefore, the uncertainty and disturbance of the system model are two inevitable factors deteriorating the control performance of quadrotor multi-UAV systems even appearing unpredictable responses, which leads to some new challenges in modeling and control of quadrotor multi-UAV systems. In this regard, there are already many adaptive or robust strategies for multi-agent systems, such as radial basis function (RBF) neural networks, adaptive compensators, and active disturbance rejection control (ADRC) [16–18].

Among them, the UDE observer method provides an effective yet powerful robust scheme using filter control design due to its advantage in handling uncertainty and disturbances simultaneously, which have been extensively used to enhance the robustness and adaptability of uncertain system models [19–21]. Similar to disturbance observer (DOB) [22], UDE can directly suppress lumped uncertainty by selecting an appropriate filter bandwidth. Compared to adaptive compensator [17] and ADRC [18], it does not require tedious adjustments of control gain and control parameters. In addition, UDE can handle time-varying external disturbances with complex structures, which RBF neural networks [16] cannot achieve, and it needs no pseudo-inverse of the nominal model, which is different from DOB [22]. In Refs. [23,24], UDE has been extended to distributed synchronization of multiple surface aircraft and consensus of a class of uncertain multi-agent systems, respectively. However, this method has not yet been considered for optimal control. Moreover, since the various high-frequency electronic signal interferences in the air, switching communication topologies frequently occur within an effective detection range of each drone [25,26]. Nevertheless, to the best of our knowledge, very little research on the optimal strategy is proposed to solve the formation control problems of quadrotor multi-UAV systems with switching communication topologies, particularly when system uncertainty and disturbance are considered simultaneously, which motivates our research in the present paper.

With the aforementioned observations and research background, and by considering underactuated features of quadrotor characterizing six degrees of freedom of position and attitude with only four motor control inputs, as well as the uncertainty and disturbance of system models, this paper aims to develop RL-based strategy for the study of the optimized robust formation control problem of quadrotor multi-UAV systems by systematically combining UDE observer method and switched sliding mode tracking control technique, in which the sliding mode tracker achieves adaptive switching of communication topology between drones through the ADT switching control method [27,28].

The main contributions of this paper in comparison with the recent works are listed below:

- (1). Although the problem of quadrotor multi-UAV systems optimal formation control by using RL-based method has been studied in [11–13]. However, the PE condition for the input signal has not been resolved yet. In this study, a unique gradient descent algorithm is proposed to train the critic and actor networks, and thereby leading to solve the corresponding HJB equations at all, where the PE condition is no longer required. This greatly reduces the computational burden.
- (2). The UDE-based control method has been extended to collaborative tracking control of multi-agent systems in Refs. [23,24], but the optimal control problem has not been considered. In this paper, UDE is first developed for optimal formation control of quadrotor multi-UAV systems. It can directly handle model uncertainty and unknown external disturbances, and only requires a filter with appropriate bandwidth, without requiring model prior information, parameter tuning, and sign functions, thus greatly ensuring the control quality of the optimal controller.
- (3). An ADT method is utilized to design the sliding mode surface describing switch topology with local communication interaction among UAVs, which can fully guarantee the adaptability and robustness of information interaction in quadrotor multi-UAV systems. This is simply different from many existing studies [11–13].

The structure of this work is listed as follows: In Section 2, the related preparatory knowledge is presented. Section 3 establishes a problem description. The specific design process of the developed controller is given in Section 4. Furthermore, the proposed algorithm is validated through simulation in Section 5. Finally, in Section 6, the conclusions of the study are summarized.

Notations: “ R ” represents the real numbers. The n -dimensional real vector can be represented as R^n . “ $\times$ ” is the multiplication sign. “ $(\cdot)^T$ ” represents the transpose of the matrix. “ $:=$ ” is a form of definition. “ $\arg$ ” refers to the angle of the plural. “ $\max$ ” and “ $\min$ ” stand for the maximum and minimum values, respectively. “ s ” is the complex variable of the complex plane in the frequency domain. “ $\otimes$ ” is the Kronecker product. The n -dimensional identity matrix is denoted as “ I_n ”. “ $\mathbf{1}_N$ ” is the N -dimensional column vector with all elements being 1. “ ∇ ” represents the shortened representation of the partial derivative. “ $\text{tr}(\cdot)$ ” implies the trace of the matrix. “ $\lambda_{\max}$ ” and “ $\lambda_{\min}$ ” represent the maximum and minimum eigenvalues, respectively. The Euclidean norm of a vector or matrix is denoted by “ $\|\cdot\|$ ”. “ $\exp(x)$ ” represents the exponential function, which is the x power of e .

2. Preliminaries

2.1. Graph theory

The information communication between N drones can be abstractly described using the topology graph $\tilde{G} = (\vartheta, \tilde{\varepsilon}, \tilde{A})$, where $\vartheta = \{1, 2, \dots, N\}$ is the set of nodes and $\tilde{\varepsilon} \in \vartheta \times \vartheta$ is the set of connected edges. The Laplace matrix $\tilde{L} = [l_{ij}]_{N \times N} \in R^{N \times N}$, to some extent, represents the overall communication situation of quadrotor multi-UAV systems, along with its corresponding adjacency matrix $\tilde{A} = [a_{ij}]_{N \times N} \in R^{N \times N}$. If $\tilde{A}$ satisfies $(j, i) \notin \tilde{\varepsilon}$, then $\tilde{A}$ can be calculated by $a_{ij} = 0$, and correspondingly, $\tilde{L}$ can be obtained by $l_{ij} = -a_{ij}, i \neq j$, in which $l_{ii} = \sum_{j=1}^N a_{ij}$. It should be noted that the UAV itself does not communicate, i.e., $a_{ii} = 0$ for $i \in \vartheta$. For the leader–follower case, a diagonal matrix $\tilde{B} = \text{diag}\{b_1, b_2, \dots, b_n\}$ is introduced to denote the information interaction between the leader and followers, where $b_i > 0$ if the follower i receives the leader’s information and $b_i = 0$ otherwise.

As stated in the literature [29], there is an assumption as shown below.

Assumption 1. If there is a virtual leader (desired time-varying trajectory) for quadrotor multi-UAV systems, then at least one agent must be connected to it, and it is required that each group graph has a directed spanning tree.

2.2. Neural network theory

Lemma 1. Neural networks are used to approximate the unknown part of the controlled object $\bar{f}(\bar{x}) : R^N \rightarrow R^n$ within a set range $\bar{\Omega}$ [15]:

$$\bar{f}(\bar{x}) = W^T \bar{F}(\bar{x}), \quad (1)$$

where $W \in R^{p \times n}$ is the constant weight, $\bar{F}(\bar{x}) = [\bar{F}_1(\bar{x}), \dots, \bar{F}_p(\bar{x})]^T \in R^p$ is the basis function vector expressed as:

$$\bar{F}_i(\bar{x}) = \exp\left(-\frac{(\bar{x} - \bar{c}_i)^T(\bar{x} - \bar{c}_i)}{2}\right), \quad i = 1, \dots, p,$$

where $\bar{c}_i = [\bar{c}_{i1}, \dots, \bar{c}_{iN}]^T \in R^N$ is the center of the receptive field.

Given the neural network's properties, an unknown ideal weight matrix W^* can be calculated through adaptive training. Consequently, Eq. (1) can be reformulated as:

$$\bar{f}(\bar{x}) = W^{*T} \bar{F}(\bar{x}) + \epsilon(\bar{x}), \quad (2)$$

where $W^* := \arg \min_W \{\sup_{\bar{x} \in \bar{\Omega}} \|\bar{f}(\bar{x}) - W^T \bar{F}(\bar{x})\|\}$, and $\epsilon(\bar{x}) \in R^n$ represents the bounded approximation error satisfied $\|\epsilon\| \leq \delta$, in which $\delta > 0$ is a constant.

2.3. ADT switching method

Consider a switching system [28], which switches continuously without generating jumps as follows:

$$\dot{Y} = \chi^{\sigma(t)}(Y), \quad (3)$$

where $\sigma(t) : [0, \infty) \rightarrow \Lambda$ is a piecewise continuous function starting from the right, $Y \in R^n$ is the state vector, and $\chi_l(\cdot)$ ($l = 1, \dots, m$, $l \in \Lambda$) is a smooth function. The symbol m denotes the number of switching subsystems.

In the following content, the symbol $\sigma(t)$ is simplified to σ for convenience. Then, a definition and a lemma related to switching control are listed below.

Definition 1 ([28]). Choose a constant M_0 . Then, there exists a switching signal σ that meets the following condition, with an ADT η_k :

$$M_\eta(T, t) \leq M_0 + \frac{T - t}{\eta_k}, \quad \forall T \geq t \geq 0, \quad (4)$$

where $T > 0$ is an arbitrary time, with the switching times $t_1, \dots, t_{M_\eta(T, 0)}$ on the interval $(0, T)$, and $M_\eta(T, t)$ expresses the number of switches that happen during the interval $[t, T)$. Let $\sigma = l_h$, if $t \in [t_h, t_{h+1})$, $l_h \neq l_{h+1}$, then l_h -th subsystem can be active, where h is a positive constant.

Lemma 2. Select $V_l(Y) = Y^T P_l Y$ for each subsystem $\dot{Y} = \chi_l(Y)$, and $P_l > 0$ is a positive definite matrix. If each subsystem of (3) is globally stable, and there exist the positive values λ_l that can form the following conditions:

$$\lambda_{\min}(P_l) \|Y(t)\|^2 \leq V_l(Y) \leq \lambda_{\max}(P_l) \|Y(t)\|^2, \quad (5)$$

and

$$\dot{V}(Y)|_l = \frac{\partial^T V_l(Y)}{\partial Y} \chi_l(Y) \leq -\lambda_l V_l(Y), \quad (6)$$

then the system (3) is globally stable with any switching ADT signal:

$$\eta_k \geq \eta_k^* = \frac{\kappa}{\lambda_0}, \quad (7)$$

and

$$\kappa = \ln v, \quad v = \max_{l \in \Lambda} \frac{\lambda_{\max}(P_l)}{\lambda_{\min}(P_l)}, \quad \lambda_0 = \min_{l \in \Lambda} \lambda_l, \quad l \in \Lambda, \quad (8)$$

where $\lambda_{\max}(P_l)$ stands for the largest eigenvalue of P_l and $\lambda_{\min}(P_l)$ stands for the least eigenvalue of P_l .

Next, other useful conclusions are introduced as follows.

Lemma 3. If the function $V(t) \in R$ [15] is time continuous, and satisfies:

- (1). The initial value $V(0)$ is always bounded.
- (2). The time derivative form $\dot{V}(t) \leq -o_1 V(t) + o_2$.

Thus, the following conclusion can be drawn:

$$V(t) \leq \exp(-o_1 t) V(0) + \frac{o_2}{o_1} (1 - \exp(-o_1 t)). \quad (9)$$

where $o_1 > 0$ and $o_2 > 0$ are constants.

Definition 2. Denote a nonlinear system $\dot{X}(t) = f(X(t), t)$ [15], where $X(t) \in R^n$. When its solution satisfies $\|X(t)\| \leq u_1$, $t > \bar{u}(u_1, X(0))$, then $X(t)$ is said to be semi-globally uniformly ultimately bounded (SGUUB), where u_1 and $\bar{u}(u_1, X(0))$ are constants, and $X(0)$ belongs to a compact set.

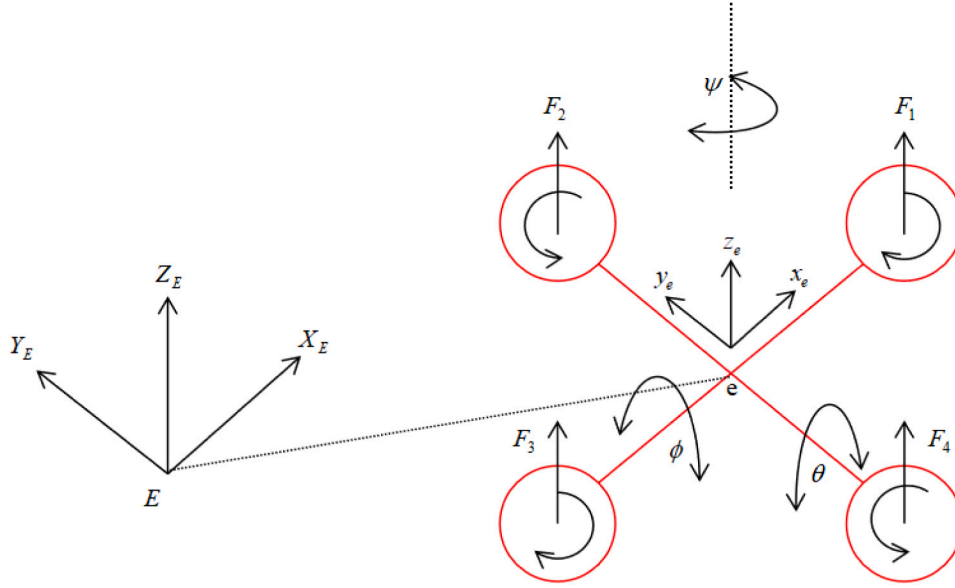


Fig. 1. Quadcopter drone configuration.

2.4. Dynamic model description of quadcopter

The following presumptions are established for the system model to streamline the quadrotor: the body is rigid and strictly symmetrical, with the origin of the body coordinate system coinciding with the center of mass and no flapping motion of the blades.

As for the body dynamics, the mathematical modeling of quadrotor is often established in two coordinate systems: the body coordinate $e = \{O_e, x_e, y_e, z_e\}$ and the world reference coordinate $E = \{O_E, X_E, Y_E, Z_E\}$, which is illustrated in Fig. 1.

There are two velocity relationships in two coordinate systems:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \bar{R}_1 \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}, \quad \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \bar{R}_2 \begin{bmatrix} w_\phi \\ w_\theta \\ w_\psi \end{bmatrix}, \quad (10)$$

where $[v_x, v_y, v_z]^T$ represent the linear velocity vectors, and $[w_\phi, w_\theta, w_\psi]^T$ represent the angular velocity vectors.

The rotation orthogonal matrixes $\bar{R}_1 \in R^{3 \times 3}$ and $\bar{R}_2 \in R^{3 \times 3}$ are used to define the position of the drone relative to a fixed inertial frame, as shown below:

$$\bar{R}_1 = \begin{bmatrix} \cos(\theta) \cos(\psi) & \sin(\phi) \sin(\theta) \cos(\psi) - \cos(\phi) \sin(\psi) & \cos(\phi) \sin(\theta) \cos(\psi) + \sin(\phi) \sin(\psi) \\ \cos(\theta) \sin(\psi) & \sin(\phi) \sin(\theta) \sin(\psi) + \cos(\phi) \cos(\psi) & \cos(\phi) \sin(\theta) \sin(\psi) + \sin(\phi) \cos(\psi) \\ -\sin(\theta) & \sin(\phi) \cos(\theta) & \cos(\phi) \cos(\theta) \end{bmatrix}, \quad (11)$$

$$\bar{R}_2 = \begin{bmatrix} 1 & \sin(\phi) \tan(\theta) & \cos(\phi) \tan(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) / \cos(\theta) & \cos(\phi) / \cos(\theta) \end{bmatrix}. \quad (12)$$

The Newton–Euler dynamic models of quadcopters are displayed as follows:

$$\begin{aligned} \mathbf{m} \times \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} &= F_f + F_g + F_d, \\ I_3 \times \begin{bmatrix} \dot{w}_\phi \\ \dot{w}_\theta \\ \dot{w}_\psi \end{bmatrix} + \begin{bmatrix} w_\phi \\ w_\theta \\ w_\psi \end{bmatrix} \times I_3 \begin{bmatrix} w_\phi \\ w_\theta \\ w_\psi \end{bmatrix} &= M_f - M_d, \end{aligned} \quad (13)$$

where $\mathbf{m}$ is the mass of the fuselage, while $F_f \in R^3$, $F_g \in R^3$, and $F_d \in R^3$ stand for the translational force, gravity, and air resistance respectively. On the other hand, the symbols $M_f \in R^{3 \times 3}$ and $M_d \in R^{3 \times 3}$ represent the rotational torque and air resistance torque respectively.

3. Problem description

Based on the model construction mentioned above, the dynamic model of the i th ($i = 1, 2, \dots, N$) UAV can be described as [30]:

$$\ddot{x}_i = \tau_{ix} - \frac{Q_{ix} \dot{x}_i}{m_i} + A_{ix}(x_i, \dot{x}_i) + d_{ix},$$

$$\begin{aligned}
\ddot{y}_i &= \tau_{iy} - \frac{Q_{iy}\dot{y}_i}{m_i} + \Delta_{iy}(y_i, \dot{y}_i) + d_{iy}, \\
\ddot{z}_i &= \tau_{iz} - \frac{Q_{iz}\dot{z}_i}{m_i} - g + \Delta_{iz}(z_i, \dot{z}_i) + d_{iz}, \\
\ddot{\phi}_i &= \tau_{i\phi} - \frac{I_i Q_{i\phi}\dot{\phi}_i}{I_{i1}} + \Delta_{i\phi}(\phi_i, \dot{\phi}_i) + d_{i\phi}, \\
\ddot{\theta}_i &= \tau_{i\theta} - \frac{I_i Q_{i\theta}\dot{\theta}_i}{I_{i2}} + \Delta_{i\theta}(\theta_i, \dot{\theta}_i) + d_{i\theta}, \\
\ddot{\psi}_i &= \tau_{i\psi} - \frac{I_i Q_{i\psi}\dot{\psi}_i}{I_{i3}} + \Delta_{i\psi}(\psi_i, \dot{\psi}_i) + d_{i\psi},
\end{aligned} \tag{14}$$

where $q_i = [x_i, y_i, z_i]^T$ is the position state vector of the body in the inertial coordinate system, $\mathbf{q}_i = [\phi_i, \theta_i, \psi_i]^T$ are the pitch, roll, and yaw angle vectors, respectively, and $\tau_{i\cdot}$ ($\cdot = \phi, \theta, \psi$) represents the control torque. $\tau_{i\cdot}$ ($\cdot = x, y, z$) can be represented as:

$$\begin{aligned}
\tau_{ix} &= (\cos(\phi_i) \sin(\theta_i) \cos(\psi_i) + \sin(\phi_i) \sin(\psi_i)) \tau_{if}, \\
\tau_{iy} &= (\cos(\phi_i) \sin(\theta_i) \cos(\psi_i) - \sin(\phi_i) \sin(\psi_i)) \tau_{if}, \\
\tau_{iz} &= (\cos(\phi_i) \cos(\theta_i)) \tau_{if}.
\end{aligned} \tag{15}$$

where $\tau_{if} \in R$ is the total thrust force. The meanings of other system parameters are represented as follows:

$$\begin{cases}
l_i \text{ is the radius length,} \\
m_i \text{ is the mass of the fuselage,} \\
g \text{ is the acceleration of gravity,} \\
I_{i1}, I_{i2}, \text{ and } I_{i3} \text{ expresses the moment of inertia around the x, y, and z axis,} \\
Q_{i\cdot} (\cdot = x, y, z, \phi, \theta, \psi) \text{ denote the drag coefficients,} \\
\Delta_{i\cdot} (\cdot = x, y, z, \phi, \theta, \psi) = [\Delta_{iq}, \Delta_{iq}]^T \text{ denote the model uncertainty,} \\
d_{i\cdot} (\cdot = x, y, z, \phi, \theta, \psi) = [d_{iq}, d_{iq}]^T \text{ denote the external disturbances.}
\end{cases}$$

The power board distributes control input commands to the four rotors to achieve power distribution of the quadcopter. The control inputs of the i th quadrotor are defined as:

$$\begin{bmatrix} \tau_{if} \\ \tau_{i\phi} \\ \tau_{i\theta} \\ \tau_{i\psi} \end{bmatrix} = \begin{bmatrix} b_1 & b_1 & b_1 & b_1 \\ -b_1 & 0 & b_1 & 0 \\ 0 & -b_1 & 0 & b_1 \\ b_2 & -b_2 & b_2 & -b_2 \end{bmatrix} \begin{bmatrix} \Omega_{i1}^2 \\ \Omega_{i2}^2 \\ \Omega_{i3}^2 \\ \Omega_{i4}^2 \end{bmatrix}, \tag{16}$$

where b_1 is the lift coefficient, b_2 is the reverse moment coefficient, and $\Omega_{i\cdot}$ ($\cdot = 1, 2, 3, 4$) represent the rotational speeds of the four propellers.

Assumption 2. The model uncertainty Δ_i and external disturbances d_i are bounded, and meet $|\Delta_i + d_i| \leq \epsilon_{1i}$, i.e., $|\dot{\Delta}_i + \dot{d}_i| \leq \epsilon_{2i}$, where $\epsilon_{1i} > 0$ and $\epsilon_{2i} > 0$ are constants.

Control objective: Our overarching control goal is to devise a control strategy that empowers the given quadrotor multi-UAV systems, with model uncertainty and unknown external disturbances, to achieve optimal formation tracking control amidst dynamic switching communication interaction networks.

4. Reinforcement learning-based formation controller design

According to model (14), the specific optimal controller design content for the position and attitude systems of i th ($i = 1, 2, \dots, N$) drone.

4.1. Position system UDE observer design

The i th UAV's position system can be written in the following form:

$$\begin{aligned}
\dot{q}_i &= v_i, \\
\dot{v}_i &= \tau_{iq} + F_{iq} + u_{iqd}.
\end{aligned} \tag{17}$$

where $v_i \in R^3$ is the linear velocity vector, $\dot{v}_i \in R^3$ denotes the acceleration, and $\tau_{iq} = [\tau_{ix}, \tau_{iy}, \tau_{iz}]^T$. $F_{iq} = [-Q_{ix}\dot{x}_i/m_i, -Q_{iy}\dot{y}_i/m_i, -Q_{iz}\dot{z}_i/m_i]^T$, and $u_{iqd} = \Delta_{iq} + d_{iq} \in R^3$ represents the lumped uncertainty.

Then, a time-varying bounded desired trajectory $q_{i0} \in R^3$ is set for q_i , and $v_{i0} \in R^3$ is the corresponding velocity vector. The state tracking error and its first time derivative form are designed as $e_{iq} = q_i - q_{i0} - h_i$ and $e_{iv} = v_i - v_{i0}$, respectively, in which $h_i \in R^3$ denotes the formation distance.

By shifting the term, Eq. (17) can be rewritten as:

$$u_{iqd} = \dot{v}_i - \tau_{iq} - F_{iq}, \tag{18}$$

and after frequency domain transformation, it can be obtained that:

$$u_{iqd}(s) = s v_i(s) - \tau_{iq}(s) - F_{iq}(s). \tag{19}$$

Design a low-pass filter $G_{fi}(s) = 1/(\rho s + 1)$, then substitute it into Eq. (19) and calculate the inverse Laplace transform. The following UDE observer can be obtained:

$$\begin{aligned}\hat{u}_{iqd} &= G_{fi}(s) (\dot{v}_i - u_{iq} - F_{iq} + \hat{u}_{iqd}) = \frac{1}{\rho s} (\dot{v}_i - u_{iq} - F_{iq} + \hat{u}_{iqd}) \\ &= \frac{1}{\rho} \left(v_i(t) - v_i(0) - \int_0^t (u_{iq}(\zeta) + F_{iq}(\zeta)) d\zeta \right), \quad t \rightarrow \infty.\end{aligned}\quad (20)$$

where the symbol ρ is the adjustable parameter, and u_{iq} is the nominal part of the controller to be designed.

If $u_{iq} \in R^3$ has stabilized the nominal system of controlled model (17), the following estimation error function can be obtained:

$$\tilde{u}_{iqd} = u_{iqd} - \hat{u}_{iqd}. \quad (21)$$

Subsequently, Eq. (21) can be frequency-domain transformed into:

$$\tilde{u}_{iqd}(s) = (1 - G_{fi}(s)) u_{iqd}(s) = \frac{\rho s}{\rho s + 1} u_{iqd}(s). \quad (22)$$

By transferring items further, it can be concluded that:

$$\rho s \tilde{u}_{iqd}(s) = -\tilde{u}_{iqd}(s) + \rho s u_{iqd}(s). \quad (23)$$

After performing the inverse transformation on Eq. (23), an important conclusion can be drawn as follows:

$$\dot{\tilde{u}}_{iqd} = -\frac{1}{\rho} \tilde{u}_{iqd} + \dot{u}_{iqd}. \quad (24)$$

4.2. Position system switched sliding mode design

Firstly, the formation tracking error vector is provided as follows:

$$E_{iq}^\sigma = \sum_{j=1}^N a_{ij}^\sigma ((q_i - h_i) - (q_j - h_j)) + b_i^\sigma (q_i - q_{i0} - h_i), \quad (25)$$

where the symbols a_{ij}^σ and b_i^σ represent the switching connectivity parameters, and Eq. (25) could be rewritten in a compact form:

$$E_q^\sigma = ((\tilde{L}^\sigma + \tilde{B}^\sigma) \otimes I_3) \times (q_i - h_i - q_{i0} \times \mathbf{1}_N), \quad (26)$$

Secondly, since Eq. (25) can be expressed in the form as:

$$E_{iq}^\sigma = \sum_{j=1}^N a_{ij}^\sigma (e_{iq} - e_{jq}) + b_i^\sigma e_{iq}, \quad (27)$$

and then the sliding mode surface can be designed:

$$y_{ip} = e_{iv} + k_{ip} e_{iq}, \quad (28)$$

where $k_{ip} > 0$ is a constant control parameter.

Denote $\tilde{L}^\sigma + \tilde{B}^\sigma = L^\sigma$, and multiplies L^σ in both sides of Eq. (28), the sliding mode error vector can be obtained as follows:

$$Y_{ip}^\sigma = \dot{E}_{iq}^\sigma + k_{ip} E_{iq}^\sigma = \sum_{j=1}^N a_{ij}^\sigma (y_{ip} - y_{jp}) + b_i^\sigma y_{ip}. \quad (29)$$

4.3. Position system optimal formation control

On the basis of the previous two subsections, in this subsection, we attempt to develop an overall optimization control framework for the i th ($i = 1, 2, \dots, N$) UAV's position system.

Consider the following performance index function combined with Eqs. (17), (20), and (29):

$$J_{ip}(Y_{ip}^\sigma) = \int_t^\infty c_{ip}(Y_{ip}^\sigma(\tau), \tau_{iq}(Y_{ip}^\sigma)) d\tau, \quad (30)$$

where $c_{ip} = Y_{ip}^{\sigma T} Y_{ip}^\sigma + \tau_{iq}^T \tau_{iq} \in R$ is the cost function. Then, the optimized performance index function based on this could be gotten as:

$$J_{ip}^*(Y_{ip}^\sigma) = \min_{\tau_{iq} \in O(\mathcal{U})} \left\{ \int_t^\infty c_{ip}(Y_{ip}^\sigma(\tau), \tau_{iq}(Y_{ip}^\sigma)) d\tau \right\} = \int_t^\infty c_{ip}(Y_{ip}^\sigma(\tau), \tau_{iq}^*(Y_{ip}^\sigma)) d\tau, \quad (31)$$

where $\mathcal{U} \in R^3$ is a given compact set, and at this point, $\tau_{iq} \in O(\mathcal{U})$ can be expressed as admissible control, if $\tau_{iq}(0) = 0$.

Taking the derivative of Eq. (31) with respect to time yields the HJB dynamic equation:

$$\begin{aligned}H_{ip}(Y_{ip}^\sigma, \tau_{iq}^*, J_{ip}^*(Y_{ip}^\sigma)) &= Y_{ip}^{\sigma T} Y_{ip}^\sigma + \tau_{iq}^{*T} \tau_{iq}^* + \frac{dJ_{ip}^*(Y_{ip}^\sigma)}{dY_{ip}^\sigma} \dot{Y}_{ip}^\sigma \\ &= Y_{ip}^{\sigma T} Y_{ip}^\sigma + \tau_{iq}^{*T} \tau_{iq}^* + \nabla^T J_{ip}^* \left(A_{i1}^\sigma (\tau_{iq}^* + F_{iq} + u_{iqd} - \dot{v}_{i0}) + f_{ip} - A_{i2}^\sigma \dot{y}_{jp} \right),\end{aligned}\quad (32)$$

where the symbols $A_{i1}^\sigma = \sum_{j=1}^N a_{ij}^\sigma + b_i^\sigma$ and $A_{i2}^\sigma = \sum_{j=1}^N a_{ij}^\sigma$, $\dot{v}_{i0} \in R^3$ represents the acceleration of leader, and $f_{ip} = A_{i1}^\sigma k_{ip} e_{iv}$. For convenience, the symbol $H_{ip}(Y_{ip}^\sigma, \tau_{iq}^*, J_{ip}^*(Y_{ip}^\sigma)) \in R^3$ is replaced by H_{ip} in the subsequent content.

Then, by solving $H_{ip} = 0$, we could obtain the following preliminary solution:

$$\tau_{iq}^* = -\frac{A_{i1}^\sigma}{2} \nabla J_{ip}^*, \quad (33)$$

and Eq. (32) can be reorganized by combining with Eq. (33) as follows:

$$H_{ip} = Y_{ip}^{\sigma T} Y_{ip}^\sigma - \frac{A_{i1}^{\sigma 2}}{4} \nabla^T J_{ip}^* \nabla J_{ip}^* + \nabla^T J_{ip}^* F_{iq} + \nabla^T J_{ip}^* u_{iqd} - \nabla^T J_{ip}^* \dot{v}_{i0} + \nabla^T J_{ip}^* f_{ip} - \nabla^T J_{ip}^* A_{i2}^\sigma \dot{y}_{jp}. \quad (34)$$

4.4. Position system critic-actor neural network algorithm

Due to the difficulty of solving the nonlinear function ∇J_{ip}^* ($i = 1, 2, \dots, N$), an RL-based algorithm is considered to handle this problem and complete the subsequent solving process.

The term ∇J_{ip}^* can be decomposed into $\nabla J_{ip}^* = 2K_{ip}Y_{ip}^\sigma + 1/A_{i1}^\sigma X_{ip}$, in which $X_{ip} = -2K_{ip}Y_{ip}^\sigma + 2\hat{u}_{iqd} + A_{i1}^\sigma \nabla J_{ip}^*$. K_{ip} is a control parameter to be designed later, and then the following result can be obtained:

$$\tau_{iq}^* = -K_{ip}Y_{ip}^\sigma + \hat{u}_{iqd} - \frac{1}{2}X_{ip}. \quad (35)$$

According to Lemma 1, X_{ip} is processed by a neural network as follows:

$$\hat{X}_{ip} = W_{ip}^{*T} S_{ip}(Y_{ip}^\sigma) + \epsilon_{ip}(Y_{ip}^\sigma), \quad (36)$$

where $W_{ip}^* \in R^{p \times 3}$ is the ideal weight, $S_{ip}(Y_{ip}^\sigma) \in R^p$ is the basis function, and $\epsilon_{ip}(Y_{ip}^\sigma) \in R^3$ stands for the bounded approximation error satisfied $\|\epsilon_{ip}(Y_{ip}^\sigma)\| \leq \delta_i$, $\delta_i > 0$. $S_{ip}(Y_{ip}^\sigma)$ and $\epsilon_{ip}(Y_{ip}^\sigma)$ are replaced by S_{ip} and ϵ_{ip} after that, for ease of writing, respectively.

Subsequently, by inserting Eq. (36) into Eq. (35) yields:

$$\hat{\tau}_{iq}^* = -K_{ip}Y_{ip}^\sigma + \hat{u}_{iqd} - \frac{1}{2}W_{ip}^{*T} S_{ip} - \frac{1}{2}\epsilon_{ip}. \quad (37)$$

Based on Eq. (37), a critic-actor RL algorithm is proposed, which is designed to satisfy the Bellman error with $H_{ip} - \hat{H}_{ip} \rightarrow 0$, where $\hat{H}_{ip}$ is the corresponding estimated state.

The nonlinear function ∇J_{ip}^* is trained by critic network as follows:

$$\nabla \hat{J}_{ip}^* = \frac{2}{A_{i1}^\sigma} K_{ip} Y_{ip}^\sigma - \frac{2}{A_{i1}^\sigma} \hat{u}_{iqd} + \frac{1}{A_{i1}^\sigma} \hat{W}_{icp}^T S_{ip}, \quad (38)$$

where $\hat{W}_{icp} \in R^{p \times 3}$ denotes the critic neural network weight, and is trained by the following update rate:

$$\dot{\hat{W}}_{icp} = -l_{icp} S_{ip} S_{ip}^T \hat{W}_{icp}, \quad (39)$$

where $l_{icp} > 0$ is a constant.

Furthermore, the optimal controller is trained by actor network in the following form:

$$\hat{\tau}_{iq}^* = -K_{ip}Y_{ip}^\sigma + \hat{u}_{iqd} - \frac{1}{2}\hat{W}_{iap}^T S_{ip}, \quad (40)$$

where $u_{iq} = -K_{ip}Y_{ip}^\sigma - \frac{1}{2}\hat{W}_{iap}^T S_{ip}$, and $\hat{W}_{iap} \in R^{p \times 3}$ is the actor neural network weight, which is trained by the following update rate:

$$\dot{\hat{W}}_{iap} = -S_{ip} S_{ip}^T (l_{iap} (\hat{W}_{iap} - \hat{W}_{icp}) + l_{icp} \hat{W}_{icp}), \quad (41)$$

where $l_{iap} > 0$ is a constant.

Remark 1. Let $Z_{ip} = (\hat{W}_{iap} - \hat{W}_{icp})^T (\hat{W}_{iap} - \hat{W}_{icp})$, and take the time derivative of it based on Eqs. (39), (41), then the result could be easily obtained as $\dot{Z}_{ip} = -\frac{l_{iap}}{2} \frac{\partial Z_{ip}}{\partial \hat{W}_{iap}} S_{ip} S_{ip}^T \frac{\partial Z_{ip}}{\partial \hat{W}_{iap}} \leq 0$. This indicates that when $Z_{ip} = 0$, the training rates (39) and (41) can make $\frac{\partial \hat{H}_{ip}}{\partial \hat{W}_{iap}} = \frac{1}{2} S_{ip} S_{ip}^T (\hat{W}_{iap} - \hat{W}_{icp}) = 0$ hold, which proves that the given RL algorithm, i.e., Eqs. (38)–(41), can meet the Bellman error optimization constraint.

Afterwards, the HJB equation can be approximated as:

$$\hat{H}_{ip} = Y_{ip}^{\sigma T} Y_{ip}^\sigma + \hat{\tau}_{iq}^{*T} \hat{\tau}_{iq}^* + \nabla^T \hat{J}_{ip}^* \left(A_{i1}^\sigma (\tau_{iq}^* + F_{iq} + u_{iqd} - \dot{v}_{i0}) + f_{ip} - A_{i2}^\sigma \dot{y}_{jp} \right). \quad (42)$$

The specific controller design scheme can be displayed in Fig. 2.

The RL-based algorithm design procedures are shown as follows:

Algorithm 1: Framework of optimal formation control method.

Initialization: Select appropriate initial conditions. The sliding mode error vector and UDE have been prepared in Eq. (20) and Eq. (29).

Process:

step 1: while $i < N$ do

step 2: build the performance index functions J_{ip} and J_{ip}^* .

step 3: design the critic-actor neural network updating laws $\dot{\hat{W}}_{icp}$ and $\dot{\hat{W}}_{iap}$.

step 4: if the Bellman error is not minimal, then the actor's weights are updated based on the HJB equation and the critic weights.

else

use the current actor neural network weight.

step 5: obtain the final solution, i.e. the optimal controller $\hat{\tau}_{iq}^*$.

step 6: end while

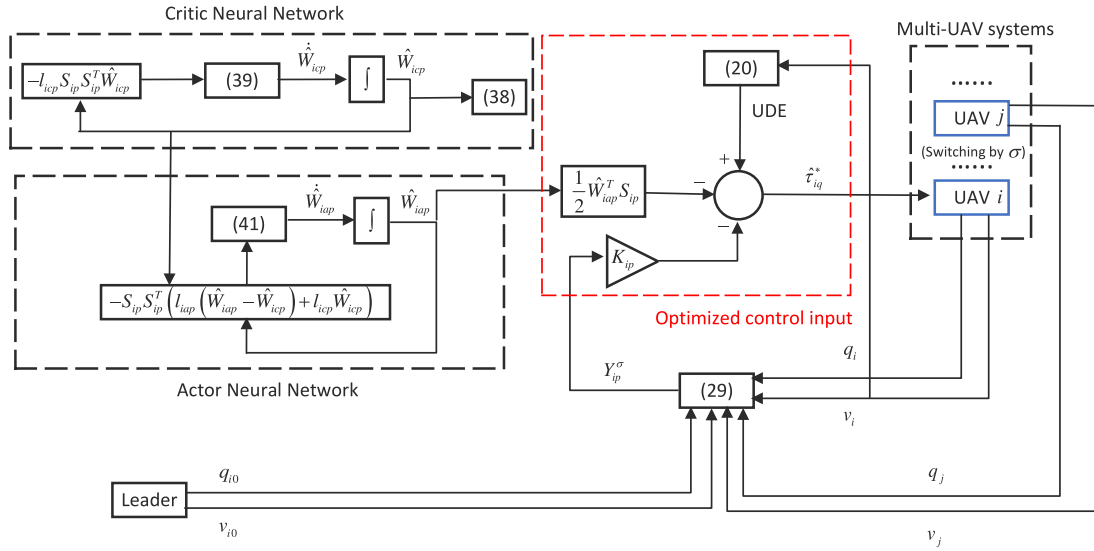


Fig. 2. The control scheme for the i th UAV position system ($i = 1, 2, \dots, N$).

4.5. Position system stability analysis

Since the stability of the controlled closed-loop system (17) needs to be verified, an important conclusion is given as follows.

Theorem 1. For the controlled system (17), if the drones are all interconnected, satisfying Assumption 1, and under the given RL-based optimization control algorithm, where includes the critic-network (38) and its update rate (39), the actor-network (40) and its update rate (41), and its control parameters meet:

$$K_{ip} > \frac{7}{4}, \quad l_{iap} > l_{icp} > \frac{l_{iap}}{2} > 0, \quad i = 1, 2, \dots, N, \quad (43)$$

and the communication topologies are randomly switched with any ADT signal as:

$$\eta_\kappa \geq \eta_\kappa^* = \frac{\kappa}{\lambda_0}, \quad (44)$$

where

$$\kappa = \ln v, \quad v = \max_{\sigma \in \Lambda} \frac{\lambda_{\max}(\bar{h}^\sigma)}{\lambda_{\min}(\bar{h}^\sigma)}, \quad \lambda_0 = \min_{\sigma \in \Lambda} \{\lambda_{0\sigma}\}, \quad \lambda_{0\sigma} > 0, \quad (45)$$

where $\lambda_{\max}(\bar{h}^\sigma)$ and $\lambda_{\min}(\bar{h}^\sigma)$ represent the maximum and minimum eigenvalues of $\bar{h}^\sigma$, which will be designed later. Then, all error signals of the position system (17) are SGUUB, which meets Definition 2. The tracking errors e_{iq} and e_{iv} can converge to bounded regions close to the origin, respectively.

Proof. Select a Lyapunov function as follows:

$$V_p^\sigma = \frac{1}{2} y_p^T (L^\sigma \otimes I_3) y_p + \frac{1}{2} \sum_{i=1}^N \text{tr}(\tilde{W}_{icp}^T \tilde{W}_{icp}) + \frac{1}{2} \sum_{i=1}^N \text{tr}(\tilde{W}_{iap}^T \tilde{W}_{iap}) + \frac{1}{2} \sum_{i=1}^N \tilde{u}_{iqd}^T \tilde{u}_{iqd}, \quad (46)$$

where $y_p = [y_{1p}, y_{2p}, \dots, y_{Np}]^T \in \mathbb{R}^{3N}$. $\tilde{W}_{icp} = \hat{W}_{icp} - W_{ip}^*$ and $\tilde{W}_{iap} = \hat{W}_{iap} - W_{ip}^*$ represent the weight estimation errors. Eq. (46) can be redesigned on the basis of Eq. (29):

$$V_p^\sigma = \frac{1}{2} Y_p^{\sigma T} (L^{\sigma^{-1}} \otimes I_3) Y_p^\sigma + \frac{1}{2} \sum_{i=1}^N \text{tr}(\tilde{W}_{icp}^T \tilde{W}_{icp}) + \frac{1}{2} \sum_{i=1}^N \text{tr}(\tilde{W}_{iap}^T \tilde{W}_{iap}) + \frac{1}{2} \sum_{i=1}^N \tilde{u}_{iqd}^T \tilde{u}_{iqd}, \quad (47)$$

where $Y_p^\sigma = [Y_{1p}^\sigma, Y_{2p}^\sigma, \dots, Y_{Np}^\sigma]^T \in \mathbb{R}^{3N}$.

Since L^σ is a positive definite matrix, the time derivative of V_p^σ by substituting Eqs. (17), (39), and (41) into Eq. (47) can be calculated as:

$$\begin{aligned} \dot{V}_p^\sigma = & \sum_{i=1}^N Y_{ip}^{\sigma T} (\tau_{iq} + F_{ieq} - \dot{v}_{i0} + u_{iqd}) - \sum_{i=1}^N \text{tr} \left[\tilde{W}_{icp}^T (l_{icp} S_{ip} S_{ip}^T \dot{\hat{W}}_{icp}) \right] \\ & - \sum_{i=1}^N \text{tr} \left[\tilde{W}_{iap}^T S_{ip} S_{ip}^T (l_{icp} \dot{\hat{W}}_{icp} + l_{iap} (\dot{\hat{W}}_{iap} - \dot{\hat{W}}_{icp})) \right] + \sum_{i=1}^N \tilde{u}_{iqd}^T \dot{\tilde{u}}_{iqd}, \end{aligned} \quad (48)$$

where the bounded function $F_{ieq} = F_{iq} + k_{ip} e_{iq}$.

Then, it can be further derived through inserting Eq. (40) into Eq. (48) as follows:

$$\dot{V}_p^\sigma = - \sum_{i=1}^N Y_{ip}^{\sigma T} K_{ip} Y_{ip}^\sigma - \frac{1}{2} \sum_{i=1}^N Y_{ip}^{\sigma T} \dot{\hat{W}}_{iap}^T S_{ip} + \sum_{i=1}^N Y_{ip}^{\sigma T} F_{ieq} - \sum_{i=1}^N Y_{ip}^{\sigma T} \dot{v}_{i0}$$

$$\begin{aligned}
& - \sum_{i=1}^N l_{icp} \text{tr} \left(\tilde{W}_{icp}^T S_{ip} S_{ip}^T \hat{W}_{icp} \right) - \sum_{i=1}^N \text{tr} \left[\tilde{W}_{iap}^T S_{ip} S_{ip}^T (l_{icp} \hat{W}_{icp} + l_{iap} (\hat{W}_{iap} - \hat{W}_{icp})) \right] \\
& + \sum_{i=1}^N Y_{ip}^{\sigma T} \tilde{u}_{iqd} + \sum_{i=1}^N \tilde{u}_{iqd}^T \dot{\tilde{u}}_{iqd}.
\end{aligned} \tag{49}$$

Subsequently, on the basis of conditions $\tilde{W}_{icp} = \hat{W}_{icp} - W_{ip}^*$ and $\tilde{W}_{iap} = \hat{W}_{iap} - W_{ip}^*$, the following facts can be held:

$$\begin{cases} \text{tr} \left(\tilde{W}_{icp}^T S_{ip} S_{ip}^T \hat{W}_{icp} \right) = \frac{1}{2} \text{tr} \left(\tilde{W}_{icp}^T S_{ip} S_{ip}^T \tilde{W}_{icp} \right) + \frac{1}{2} \text{tr} \left(\hat{W}_{icp}^T S_{ip} S_{ip}^T \hat{W}_{icp} \right) - \frac{1}{2} \text{tr} \left(W_{ip}^{*T} S_{ip} S_{ip}^T W_{ip}^* \right), \\ \text{tr} \left(\tilde{W}_{iap}^T S_{ip} S_{ip}^T \hat{W}_{iap} \right) = \frac{1}{2} \text{tr} \left(\tilde{W}_{iap}^T S_{ip} S_{ip}^T \tilde{W}_{iap} \right) + \frac{1}{2} \text{tr} \left(\hat{W}_{iap}^T S_{ip} S_{ip}^T \hat{W}_{iap} \right) - \frac{1}{2} \text{tr} \left(W_{ip}^{*T} S_{ip} S_{ip}^T W_{ip}^* \right). \end{cases} \tag{50}$$

Furthermore, the stability analysis could be continued by substituting Eqs. (50) to (49):

$$\begin{aligned}
\dot{V}_p^\sigma & = - \sum_{i=1}^N Y_{ip}^{\sigma T} K_{ip} Y_{ip}^\sigma + \sum_{i=1}^N Y_{ip}^{\sigma T} F_{ieq} - \sum_{i=1}^N Y_{ip}^{\sigma T} \dot{v}_{i0} - \frac{1}{2} \sum_{i=1}^N Y_{ip}^{\sigma T} \hat{W}_{iap}^T S_{ip} \\
& - \frac{1}{2} \sum_{i=1}^N l_{icp} \text{tr} \left(\tilde{W}_{icp}^T S_{ip} S_{ip}^T \tilde{W}_{icp} \right) - \frac{1}{2} \sum_{i=1}^N l_{icp} \text{tr} \left(\hat{W}_{icp}^T S_{ip} S_{ip}^T \hat{W}_{icp} \right) \\
& - \frac{1}{2} \sum_{i=1}^N l_{iap} \text{tr} \left(\tilde{W}_{iap}^T S_{ip} S_{ip}^T \tilde{W}_{iap} \right) - \frac{1}{2} \sum_{i=1}^N l_{iap} \text{tr} \left(\hat{W}_{iap}^T S_{ip} S_{ip}^T \hat{W}_{iap} \right) \\
& - \sum_{i=1}^N l_{icp} \text{tr} \left(\tilde{W}_{iap}^T S_{ip} S_{ip}^T \hat{W}_{icp} \right) + \sum_{i=1}^N l_{iap} \text{tr} \left(\tilde{W}_{icp}^T S_{ip} S_{ip}^T \hat{W}_{icp} \right) \\
& + \sum_{i=1}^N W_{ip}^{*T} S_{ip} S_{ip}^T W_{ip}^* + \sum_{i=1}^N Y_{ip}^{\sigma T} \tilde{u}_{iqd} + \sum_{i=1}^N \tilde{u}_{iqd}^T \dot{\tilde{u}}_{iqd}.
\end{aligned} \tag{51}$$

By utilizing the conclusion (24) and Young's inequality ($\bar{a}\bar{b} \leq \frac{1}{2}\bar{a}^2 + \frac{1}{2}\bar{b}^2$), there are the following inequalities:

$$\begin{cases} Y_{ip}^{\sigma T} F_{ieq} \leq \frac{1}{2} \|Y_{ip}^\sigma\|^2 + \frac{1}{2} \|F_{ieq}\|^2, & -Y_{ip}^{\sigma T} \dot{v}_{i0} \leq \frac{1}{2} \|Y_{ip}^\sigma\|^2 + \frac{1}{2} \|\dot{v}_{i0}\|^2, \\ -Y_{ip}^{\sigma T} \dot{v}_{i0} \leq \frac{1}{2} \|Y_{ip}^\sigma\|^2 + \frac{1}{2} \|\dot{v}_{i0}\|^2, & Y_{ip}^{\sigma T} \tilde{u}_{iqd} \leq \frac{1}{2} \|Y_{ip}^\sigma\|^2 + \frac{1}{2} \|\tilde{u}_{iqd}\|^2, \\ -\frac{1}{2} Y_{ip}^{\sigma T} \hat{W}_{iap}^T S_{ip} \leq \frac{1}{4} \|Y_{ip}^\sigma\|^2 + \frac{1}{4} \hat{W}_{iap}^T S_{ip} S_{ip}^T \hat{W}_{iap}, \\ \tilde{u}_{iqd}^T \dot{\tilde{u}}_{iqd} \leq \beta_{1i} \tilde{u}_{iqd}^T \tilde{u}_{iqd} + \frac{\epsilon_{2i}^2}{4\beta_{1i}} \leq -\tilde{u}_{iqd}^T \frac{1}{\rho} \tilde{u}_{iqd} + \beta_{1i} \tilde{u}_{iqd}^T \tilde{u}_{iqd} + \frac{\epsilon_{2i}^2}{4\beta_{1i}}, \\ (l_{iap} - l_{icp}) \text{tr} \left(\tilde{W}_{iap}^T S_{ip} S_{ip}^T \tilde{W}_{icp} \right) \leq \frac{l_{iap} - l_{icp}}{2} \text{tr} \left(\tilde{W}_{iap}^T S_{ip} S_{ip}^T \tilde{W}_{iap} \right) + \frac{l_{iap} - l_{icp}}{2} \text{tr} \left(\hat{W}_{icp}^T S_{ip} S_{ip}^T \hat{W}_{icp} \right), \end{cases} \tag{52}$$

where $\beta_{1i} > 0$ is a constant, and then inserting the above seven inequalities into Eq. (51) yields:

$$\begin{aligned}
\dot{V}_p^\sigma & \leq - \sum_{i=1}^N \left(K_{ip} - \frac{7}{4} \right) \|Y_{ip}^\sigma\|^2 - \sum_{i=1}^N \left(\frac{l_{iap}}{2} - \frac{1}{4} \right) \text{tr} \left(\hat{W}_{iap}^T S_{ip} S_{ip}^T \hat{W}_{iap} \right) - \sum_{i=1}^N \frac{l_{icp}}{2} \text{tr} \left(\tilde{W}_{icp}^T S_{ip} S_{ip}^T \tilde{W}_{icp} \right) \\
& - \sum_{i=1}^N \frac{l_{icp}}{2} \text{tr} \left(\tilde{W}_{iap}^T S_{ip} S_{ip}^T \tilde{W}_{iap} \right) - \sum_{i=1}^N \left(l_{icp} - \frac{l_{iap}}{2} \right) \text{tr} \left(\hat{W}_{icp}^T S_{ip} S_{ip}^T \hat{W}_{icp} \right) + \sum_{i=1}^N \|W_{ip}^* S_{ip}\|^2 \\
& + \frac{1}{2} \sum_{i=1}^N \|F_{ieq}\|^2 + \frac{1}{2} \sum_{i=1}^N \|\dot{v}_{i0}\|^2 + \frac{1}{2} \sum_{i=1}^N \|\tilde{u}_{iqd}\|^2 + \sum_{i=1}^N \left(-\tilde{u}_{iqd}^T \frac{1}{\rho} \tilde{u}_{iqd} + \beta_{1i} \tilde{u}_{iqd}^T \tilde{u}_{iqd} + \frac{\epsilon_{2i}^2}{4\beta_{1i}} \right).
\end{aligned} \tag{53}$$

Remark 2. According to Lemma 3, the robust control term can be further simplified as $-\tilde{u}_{iqd}^T \frac{1}{\rho} \tilde{u}_{iqd} + \beta_{1i} \tilde{u}_{iqd}^T \tilde{u}_{iqd} + \frac{\epsilon_{2i}^2}{4\beta_{1i}} \leq -2\epsilon_{3i} \left(\frac{1}{2} \tilde{u}_{iqd}^T \tilde{u}_{iqd} \right) + \gamma_{1i}$, in which $\epsilon_{3i} = \lambda_{\min}(1/\rho) - \beta_{1i} > 0$, $\gamma_{1i} = \epsilon_{2i}^2/(4\beta_{1i}) > 0$.

According to Eq. (43) and Remark 2, the above inequality can be further written as follows:

$$\dot{V}_p^\sigma \leq - \sum_{i=1}^N \left(K_{ip} - \frac{7}{4} \right) \|Y_{ip}^\sigma\|^2 - \sum_{i=1}^N \frac{l_{icp}}{2} \lambda_{\min}(S_{ip}) \text{tr} \left(\tilde{W}_{icp}^T \tilde{W}_{icp} \right) - \sum_{i=1}^N \frac{l_{icp}}{2} \lambda_{\min}(S_{ip}) \text{tr} \left(\tilde{W}_{iap}^T \tilde{W}_{iap} \right) + \vartheta_p, \tag{54}$$

where $\vartheta_p = \sum_{i=1}^N \epsilon_{ip}$, and $\epsilon_{ip} \geq \sum_{i=1}^N \|W_{ip}^* S_{ip}\|^2 + \frac{1}{2} \sum_{i=1}^N \|F_{ieq}\|^2 + \frac{1}{2} \sum_{i=1}^N \|\dot{v}_{i0}\|^2 + \gamma_{1i}$.

If the following set form can be used:

$$\alpha_p = \sum_{i=1}^N \alpha_{ip} = \sum_{i=1}^N \min \{ 2(K_{ip} - 7/4), l_{icp} \lambda_{\min}(S_{ip}), l_{icp} \lambda_{\min}(S_{ip}), 2\epsilon_{3i} \},$$

then we get:

$$\dot{V}_p^\sigma \leq -\alpha_p V_p^\sigma + \vartheta_p, \tag{55}$$

By using Lemma 3, it can continue to be written as:

$$V_p^\sigma(t) \leq \exp(-\alpha_p t) V_p^\sigma(0) + \frac{\beta_p}{\alpha_p} (1 - \exp(-\alpha_p t)). \quad (56)$$

Let $L^{\sigma^{-1}} \otimes I_3 = \bar{h}^\sigma$, based on result (54) and Lemma 2, it can be inferred that the function $V_1^\sigma = \frac{1}{2} Y_p^{\sigma T} \bar{h}^\sigma Y_p^\sigma$ satisfies the following inequality constraint:

$$\frac{1}{2} \lambda_{\min}(\bar{h}^\sigma) \|Y_p^\sigma\|^2 \leq V_1^\sigma \leq \frac{1}{2} \lambda_{\max}(\bar{h}^\sigma) \|Y_p^\sigma\|^2, \quad \sigma \in \Lambda. \quad (57)$$

And then, the following relationship can be concluded by taking the time derivative of V_1^σ :

$$\dot{V}_1^\sigma \leq -\frac{1}{2} \lambda_{0\sigma} Y_p^{\sigma T} \bar{h}^\sigma Y_p^\sigma = -\lambda_{0\sigma} V_1^\sigma.$$

Use $t_1, t_2, \dots, t_h$ to represent the time points when switching occurs, and c_h for the value of $\eta_\kappa(t)$ on $[t_h, t_{h+1})$, we can further obtain:

$$\ln V_{1c_h} \left(Y_{ph}^\sigma \right) - \ln V_{1c_h} \left(Y_{p(h-1)}^\sigma \right) \leq -\lambda_{0c_h} \eta_h, \quad (58)$$

where

$$V_{1c_h} \left(Y_{ph}^\sigma \right) \leq \exp(-\lambda_{0c_h} \eta_h) V_{1c_h} \left(Y_{p(h-1)}^\sigma \right). \quad (59)$$

Based on the above results, the Eq. (57) can be expressed as:

$$\begin{aligned} \frac{1}{2} \lambda_{\min}(\bar{h}^\sigma) \|Y_{ph}^\sigma\|^2 &\leq V_{1c_h} \left(Y_{ph}^\sigma \right) \leq \exp(-\lambda_{0c_h} \eta_h) V_{1c_h} \left(Y_{p(h-1)}^\sigma \right) \leq \exp(-\lambda_{0c_h} \eta_h) \frac{1}{2} \lambda_{\max}(\bar{h}^\sigma) \|Y_{p(h-1)}^\sigma\|^2 \\ &\leq \exp(-\lambda_{0c_h} \eta_h) \frac{\frac{1}{2} \lambda_{\max}(\bar{h}^\sigma)}{\frac{1}{2} \lambda_{\min}(\bar{h}^\sigma)} \frac{1}{2} \lambda_{\min}(\bar{h}^\sigma) \|Y_{p(h-1)}^\sigma\|^2 \leq \nu \exp(-\lambda_{0c_h} \eta_h) \frac{1}{2} \lambda_{\min}(\bar{h}^\sigma) \|Y_{p(h-1)}^\sigma\|^2. \end{aligned} \quad (60)$$

For any time t meeting $t_0 < t_1 < \dots < t_l < t < t_{l+1}$, there is an inequality by using Definition 1:

$$\begin{aligned} \frac{1}{2} \lambda_{\min}(\bar{h}^\sigma) \|Y_{pt}^\sigma\|^2 &\leq \nu \exp(-\lambda_{0c_{l+1}}(t - t_l)) \frac{1}{2} \lambda_{\min}(\bar{h}^\sigma) \|Y_{pt_l}^\sigma\|^2 \\ &\vdots \\ &\leq \nu^{l+1} \exp(-\lambda_0(t - T)) \frac{1}{2} \lambda_{\min}(\bar{h}^\sigma) \|Y_{p0}^\sigma\|^2 \\ &= \exp(\kappa(l+1) - \lambda_0(t - T)) \frac{1}{2} \lambda_{\min}(\bar{h}^\sigma) \|Y_{p0}^\sigma\|^2 \\ &= \exp(\kappa) \exp(\kappa \mathbf{M}_\eta(T, t) - \lambda_0(t - T)) \lambda_{\min}(\bar{h}^\sigma) \|Y_{p0}^\sigma\|^2, \end{aligned} \quad (61)$$

where $\lambda_0 = \min\{\lambda_{0\sigma}\}$, $\sigma \in \Lambda$.

If $\nu = 1$, $\kappa = 0$, then it can be obtained that:

$$\|Y_{pt}^\sigma\|^2 \leq \exp(-\lambda_0(t - T)) \|Y_{p0}^\sigma\|^2, \quad (62)$$

and if $\kappa > 0$, $\eta_\kappa > \kappa/\lambda_0$, i.e.,

$$\kappa \mathbf{M}_\eta(T, t) - \lambda_0(t - T) \leq \kappa \mathbf{M}_0 + \left(\frac{\kappa}{\eta_\kappa} - \lambda_0 \right) (t - T), \quad (63)$$

then an another result can be gotten by substituting Eq. (63) into Eq. (61):

$$\|Y_{pt}^\sigma\|^2 \leq \exp(\kappa) \exp(\kappa \mathbf{M}_0) \exp((\kappa/\eta_\kappa - \lambda_0)(t - T)) \|Y_{p0}^\sigma\|^2. \quad (64)$$

According to the results (56) and (64), it can be concluded that any error signal of the system (17) is SGUUB. Furthermore, the vector q_i can achieve tracking control for q_{i0} , and the same result is achieved for velocity tracking error. Therefore, Theorem 1 has been proven.

4.6. Attitude system reinforcement learning-based formation controller design

Remark 3. In fact, according to Eqs. (14) and (15), it is not difficult to see that the quadrotor system is an underactuated system with four inputs, τ_{if} , $\tau_{i\phi}$, $\tau_{i\theta}$, and $\tau_{i\psi}$, and six outputs x_i , y_i , z_i , ϕ_i , θ_i , and ψ_i , making it impossible to achieve complete control. Therefore, the design of the optimal attitude control protocol for UAVs focuses on considering yaw angle, while the rest can be synchronized with the desired state.

It should be noted that the i th ($i = 1, 2, \dots, N$) UAV's reference angle command must follow the constraints shown below :

$$\begin{aligned} \phi_{i0} &= \arcsin \left(\frac{m_i \tau_{ix} \sin(\psi_{i0}) - m_i \tau_{iy} \cos(\psi_{i0})}{\tau_{if}} \right), \\ \theta_{i0} &= \arctan \left(\frac{\tau_{ix} \cos(\psi_{i0}) + \tau_{iy} \sin(\psi_{i0})}{\tau_{iz}} \right). \end{aligned} \quad (65)$$

The second-order attitude control structure based on Eq. (65), is designed as:

$$\begin{aligned} \dot{\mathbf{q}}_i &= \mathbf{w}_i \\ \dot{\mathbf{w}}_i &= \tau_{iq} + F_{iq} + u_{iqd}, \end{aligned} \quad (66)$$

where $\mathbf{w}_i = [w_{i\phi}, w_{i\theta}, w_{i\psi}]^T$ is the angular velocity vector, $\dot{\mathbf{w}}_i = [\dot{w}_{i\phi}, \dot{w}_{i\theta}, \dot{w}_{i\psi}]^T$ represents the acceleration, and $\tau_{iq} = [\tau_{i\phi}, \tau_{i\theta}, \tau_{i\psi}]^T$. $u_{iqd} = \Delta_{iq} + d_{iq} \in R^3$ stands for the lumped uncertainty, and $F_{iq} = [F_{i\phi}, F_{i\theta}, F_{i\psi}]^T = [-l_i Q_{i\phi} \phi_i / I_{i1}, -l_i Q_{i\theta} \theta_i / I_{i2}, -l_i Q_{i\psi} \psi_i / I_{i3}]^T$.

Define the tracking errors $e_{iq} = [e_{i\phi}, e_{i\theta}, e_{i\psi}]^T = \mathbf{q}_i - \mathbf{q}_{i0}$ and $e_{iw} = [e_{iw\phi}, e_{iw\theta}, e_{iw\psi}]^T = w_i - w_{i0}$, in which $\mathbf{q}_{i0} = [\phi_{i0}, \theta_{i0}, \psi_{i0}]^T$ and $w_{i0} = [w_{i0\phi}, w_{i0\theta}, w_{i0\psi}]^T$ represent the reference command and corresponding velocity, respectively.

Because the inner-loop control of the quadrotor multi-UAV systems is similar to the outer-loop, we simplify the controller design process next. Firstly, the UDE observer is designed as follows:

$$\begin{aligned}\hat{u}_{iqd} &= [\hat{u}_{i\phi d}, \hat{u}_{i\theta d}, \hat{u}_{i\psi d}]^T = G_{fi}(s) (\dot{w}_i - u_{iq} - F_{iq} + \hat{u}_{iqd}) = \frac{1}{\rho s} (\dot{w}_i - u_{iq} - F_{iq} + \hat{u}_{iqd}) \\ &= \frac{1}{\rho} \left(w_i(t) - w_i(0) - \int_0^t (u_{iq}(\zeta) + F_{iq}(\zeta)) d\zeta \right), \quad t \rightarrow \infty,\end{aligned}\quad (67)$$

where $u_{iq} = [u_{i\phi}, u_{i\theta}, u_{i\psi}]^T$ denotes the nominal part of the controller to be designed.

Secondly, the sliding mode surfaces are constructed:

$$y_{ia} = [y_{i\phi}, y_{i\theta}, y_{i\psi}]^T = e_{iw} + k_{ia} e_{iq}, \quad (68)$$

where $k_{ia} > 0$ is a tuning parameter.

Since the roll and pitch angles need to be synchronized with the leader's information, the corresponding control inputs can be designed in the following form:

$$\begin{aligned}\tau_{i\phi} &= -K_{ia} y_{i\phi} + \hat{u}_{i\phi d}, \\ \tau_{i\theta} &= -K_{ia} y_{i\theta} + \hat{u}_{i\theta d}.\end{aligned}\quad (69)$$

where K_{ia} is a parameter to be designed later, $u_{i\phi} = -K_{ia} y_{i\phi}$, and $u_{i\theta} = -K_{ia} y_{i\theta}$.

Then, the yaw angle sliding mode error vector is designed as follows:

$$Y_{i\psi}^\sigma = \sum_{j=1}^N a_{ij}^\sigma (y_{i\psi} - y_{j\psi}) + b_i^\sigma y_{i\psi}. \quad (70)$$

Following that, we develop the performance index function as follows:

$$J_{i\psi}^\sigma(Y_{i\psi}^\sigma) = \min_{\tau_{i\psi} \in \bar{O}(\bar{\mathcal{O}})} \left\{ \int_t^\infty c_{i\psi}(Y_{i\psi}^\sigma(\tau), \tau_{i\psi}(Y_{i\psi}^\sigma)) d\tau = \int_t^\infty c_{i\psi}(Y_{i\psi}^\sigma(\tau), \tau_{i\psi}^*(Y_{i\psi}^\sigma)) d\tau, \right. \quad (71)$$

where $c_{i\psi} = Y_{i\psi}^{\sigma T} Y_{i\psi}^\sigma + \tau_{i\psi}^T \tau_{i\psi} \in \mathbb{R}$ is the cost function, and $\bar{\mathcal{O}} \in \mathbb{R}$ is a given compact set, at this point, $\tau_{i\psi} \in \bar{O}(\bar{\mathcal{O}})$ can be considered as admissible control, if $\tau_{i\psi}(0) = 0$.

Take the time derivative of Eq. (71), and it results in the following HJB equation:

$$H_{i\psi} = Y_{i\psi}^{\sigma T} Y_{i\psi}^\sigma + \tau_{i\psi}^{*T} \tau_{i\psi}^* + \frac{dJ_{i\psi}^*}{dY_{i\psi}^\sigma} \dot{Y}_{i\psi}^\sigma. \quad (72)$$

By solving $H_{i\psi} = 0$ preliminarily, we get $\tau_{i\psi}^* = -(A_{i1}^\sigma/2) \nabla J_{i\psi}^*$, and then substituting the initial solution into Eq. (72), and the HJB equation is updated to:

$$H_{i\psi} = Y_{i\psi}^{\sigma T} Y_{i\psi}^\sigma - \frac{A_{i1}^{\sigma 2}}{4} \nabla^T J_{i\psi}^* \nabla J_{i\psi}^* + \nabla^T J_{i\psi}^* F_{i\psi} + \nabla^T J_{i\psi}^* u_{i\psi d} - \nabla^T J_{i\psi}^* \dot{w}_{i0\psi} + \nabla^T J_{i\psi}^* f_{i\psi} - \nabla^T J_{i\psi}^* A_{i2}^\sigma \dot{y}_{j\psi}, \quad (73)$$

where $f_{i\psi} = A_{i1}^\sigma k_{ia} e_{iw\psi}$, and $\dot{w}_{i0\psi} \in \mathbb{R}^3$ is the acceleration of leader.

In accordance with Eqs. (35)–(41), the RL-based control framework is then established.

The nonlinear function $\nabla J_{i\psi}^*$ is trained by critic network as:

$$\nabla \hat{J}_{i\psi}^* = \frac{2}{A_{i1}^\sigma} K_{ia} Y_{i\psi}^\sigma - \frac{2}{A_{i1}^\sigma} \hat{u}_{i\psi d} + \frac{1}{A_{i1}^\sigma} \hat{W}_{ic\psi}^T S_{i\psi}, \quad (74)$$

where $\hat{W}_{ic\psi} \in \mathbb{R}^p$ is the critic neural network weight, and is trained by the following update rate:

$$\dot{\hat{W}}_{ic\psi} = -l_{ic\psi} S_{i\psi} S_{i\psi}^T \hat{W}_{ic\psi}, \quad (75)$$

where $l_{ic\psi} > 0$ is a tuning parameter.

The actor network is designed as:

$$\hat{\tau}_{i\psi}^* = -K_{ia} Y_{i\psi}^\sigma + \hat{u}_{i\psi d} - \frac{1}{2} \hat{W}_{ia\psi}^T S_{i\psi}, \quad (76)$$

where $u_{i\psi} = -K_{ia} Y_{i\psi}^\sigma - \frac{1}{2} \hat{W}_{ia\psi}^T S_{i\psi}$, and $\hat{W}_{ia\psi} \in \mathbb{R}^p$ is the actor neural network weight, which is trained by the following update rate:

$$\dot{\hat{W}}_{ia\psi} = -S_{i\psi} S_{i\psi}^T (l_{ia\psi} (\hat{W}_{ia\psi} - \hat{W}_{ic\psi}) + l_{ic\psi} \hat{W}_{ic\psi}), \quad (77)$$

where $l_{ia\psi} > 0$ is a tuning parameter.

Subsequently, we can obtain the estimated form of the HJB equation:

$$\hat{H}_{i\psi} = Y_{i\psi}^{\sigma T} Y_{i\psi}^\sigma + \hat{\tau}_{i\psi}^{*T} \hat{\tau}_{i\psi}^* + \frac{d\hat{J}_{i\psi}^*}{dY_{i\psi}^\sigma} \dot{Y}_{i\psi}^\sigma. \quad (78)$$

Due to the design process of the control algorithm for the attitude system being similar to that of the position system, the related visualization work is omitted.

4.7. Attitude system stability analysis

Remark 4. The selection of parameters is limited, and K_{ia} should be significantly larger than K_{ip} because the motion characteristics of quadcopters require attitude control to converge faster than position control. On this basis, the following important theorem is given.

Theorem 2. For the controlled system (66), if the interaction between drones satisfies Assumption 1, and under the given RL-based optimal control algorithm, where includes the critic-network (74) and its update rate (75), the actor-network (76) and its update rate (77), and its control parameters meet:

$$K_{ia} \gg K_{ip}, l_{ia\psi} > l_{ic\psi} > \frac{l_{ia\psi}}{2} > 0, i = 1, 2, \dots, N, \quad (79)$$

and the communication topology can be randomly switched with any ADT signal as:

$$\eta_{\kappa_1} \geq \eta_{\kappa_1}^* = \frac{\kappa_1}{\lambda_1}, \quad (80)$$

where

$$\kappa = \ln v, v = \max_{\sigma \in A} \frac{\lambda_{\max}(h_1^\sigma)}{\lambda_{\min}(h_1^\sigma)}, \lambda_1 = \min_{\sigma \in A} \lambda_1^\sigma, \lambda_1^\sigma > 0, \quad (81)$$

where $\lambda_{\max}(h_1^\sigma)$ and $\lambda_{\min}(h_1^\sigma)$ refer to the maximum and minimum eigenvalues of the matrix h_1^σ respectively, which will be developed later. e_{iq} and e_{iu} can approach the origin within a small bounded area, and all error signals of system (66) are SGUUB.

Proof. Choose a Lyapunov function as:

$$V_\psi^\sigma = \frac{1}{2} y_\psi^T L^\sigma y_\psi + \frac{1}{2} \sum_{i=1}^N \text{tr} \left(\tilde{W}_{ic\psi}^T \tilde{W}_{ic\psi} \right) + \frac{1}{2} \sum_{i=1}^N \text{tr} \left(\tilde{W}_{ia\psi}^T \tilde{W}_{ia\psi} \right) + \frac{1}{2} \sum_{i=1}^N \tilde{u}_{i\psi d}^T \tilde{u}_{i\psi d}, \quad (82)$$

where $y_\psi = [y_{1\psi}, y_{2\psi}, \dots, y_{N\psi}]^T \in \mathbb{R}^N$. $\tilde{W}_{ic\psi} = \hat{W}_{ic\psi} - W_{ic\psi}^*$ and $\tilde{W}_{ia\psi} = \hat{W}_{ia\psi} - W_{ia\psi}^*$ are the weight estimation errors. $\tilde{u}_{i\psi d} = u_{i\psi d} - \hat{u}_{i\psi d}$. Substituting Eq. (70) into Eq. (82) yields:

$$V_\psi^\sigma = \frac{1}{2} Y_\psi^{\sigma T} (L^{\sigma^{-1}}) Y_\psi^\sigma + \frac{1}{2} \sum_{i=1}^N \text{tr} \left(\tilde{W}_{ic\psi}^T \tilde{W}_{ic\psi} \right) + \frac{1}{2} \sum_{i=1}^N \text{tr} \left(\tilde{W}_{ia\psi}^T \tilde{W}_{ia\psi} \right) + \frac{1}{2} \sum_{i=1}^N \tilde{u}_{i\psi d}^T \tilde{u}_{i\psi d}, \quad (83)$$

where $Y_\psi^\sigma = [Y_{1\psi}^\sigma, Y_{2\psi}^\sigma, \dots, Y_{N\psi}^\sigma]^T \in \mathbb{R}^N$.

Taking the derivative of V_ψ^σ with respect to time, by inserting Eqs. (66), (75), and (77) into Eq. (83) yields:

$$\begin{aligned} \dot{V}_\psi^\sigma &= \sum_{i=1}^N Y_{i\psi}^{\sigma T} (\tau_{i\psi} + F_{ie\psi} + u_{i\psi d} - \dot{u}_{i0\psi}) - \sum_{i=1}^N \text{tr} \left[\tilde{W}_{ic\psi}^T (l_{ic\psi} S_{i\psi} S_{i\psi}^T \hat{W}_{ic\psi}) \right] \\ &\quad - \sum_{i=1}^N \text{tr} \left[\tilde{W}_{ia\psi}^T S_{i\psi} S_{i\psi}^T (l_{ic\psi} \hat{W}_{ic\psi} + l_{ia\psi} (\hat{W}_{ia\psi} - \hat{W}_{ic\psi})) \right] + \sum_{i=1}^N \tilde{u}_{i\psi d}^T \dot{\tilde{u}}_{i\psi d}, \end{aligned} \quad (84)$$

where the bounded function $F_{ie\psi} = F_{i\psi} + k_{ia} e_{i\psi}$.

Insert Eq. (76) into Eq. (84), then Eq. (84) can be further written as:

$$\begin{aligned} \dot{V}_\psi^\sigma &= - \sum_{i=1}^N Y_{i\psi}^{\sigma T} K_{ia} Y_{i\psi}^\sigma - \frac{1}{2} \sum_{i=1}^N Y_{i\psi}^{\sigma T} \hat{W}_{ia\psi}^T S_{i\psi} + \sum_{i=1}^N Y_{i\psi}^{\sigma T} F_{ie\psi} + \sum_{i=1}^N Y_{i\psi}^{\sigma T} \tilde{u}_{i\psi d} - \sum_{i=1}^N Y_{i\psi}^{\sigma T} \dot{u}_{i0\psi} \\ &\quad - \sum_{i=1}^N l_{ic\psi} \text{tr} \left(\tilde{W}_{ic\psi}^T S_{i\psi} S_{i\psi}^T \hat{W}_{ic\psi} \right) - \sum_{i=1}^N \text{tr} \left[\tilde{W}_{ia\psi}^T S_{i\psi} S_{i\psi}^T (l_{ic\psi} \hat{W}_{ic\psi} + l_{ia\psi} (\hat{W}_{ia\psi} - \hat{W}_{ic\psi})) \right] + \sum_{i=1}^N \tilde{u}_{i\psi d}^T \dot{\tilde{u}}_{i\psi d}. \end{aligned} \quad (85)$$

Following that, by referring to the existing results (21)–(24), (50), and (52), the above equation can be written as the following inequality:

$$\begin{aligned} \dot{V}_\psi^\sigma &\leq - \sum_{i=1}^N \left(K_{ia} - \frac{7}{4} \right) \|Y_{i\psi}^\sigma\|^2 - \sum_{i=1}^N \left(\frac{l_{ia\psi}}{2} - \frac{1}{4} \right) \text{tr} \left(\hat{W}_{ia\psi}^T S_{i\psi} S_{i\psi}^T \hat{W}_{ia\psi} \right) - \sum_{i=1}^N \frac{l_{ic\psi}}{2} \text{tr} \left(\tilde{W}_{ic\psi}^T S_{i\psi} S_{i\psi}^T \tilde{W}_{ic\psi} \right) \\ &\quad - \sum_{i=1}^N \frac{l_{ic\psi}}{2} \text{tr} \left(\tilde{W}_{ia\psi}^T S_{i\psi} S_{i\psi}^T \tilde{W}_{ia\psi} \right) - \sum_{i=1}^N \left(l_{ic\psi} - \frac{l_{ia\psi}}{2} \right) \text{tr} \left(\hat{W}_{ic\psi}^T S_{i\psi} S_{i\psi}^T \hat{W}_{ic\psi} \right) + \sum_{i=1}^N \|W_{i\psi}^* S_{i\psi}\|^2 \\ &\quad + \frac{1}{2} \sum_{i=1}^N \|F_{ie\psi}\|^2 + \frac{1}{2} \sum_{i=1}^N \|\dot{u}_{i0\psi}\|^2 + \sum_{i=1}^N \left(-\tilde{u}_{i\psi d}^T \frac{1}{\rho} \tilde{u}_{i\psi d} + \beta_{1i} \tilde{u}_{i\psi d}^T \tilde{u}_{i\psi d} + \frac{\epsilon_{2i}^2}{4\beta_{1i}} \right). \end{aligned} \quad (86)$$

The inequality (86) can be rewritten by considering Eq. (79) and Remark 2 as shown below:

$$\dot{V}_\psi^\sigma \leq - \sum_{i=1}^N \left(K_{ia} - \frac{7}{4} \right) \|Y_{i\psi}^\sigma\|^2 - \sum_{i=1}^N \frac{l_{ic\psi}}{2} \lambda_{\min}(S_{i\psi}) \text{tr} \left(\tilde{W}_{ic\psi}^T \tilde{W}_{ic\psi} \right) - \sum_{i=1}^N \frac{l_{ic\psi}}{2} \lambda_{\min}(S_{i\psi}) \text{tr} \left(\tilde{W}_{ia\psi}^T \tilde{W}_{ia\psi} \right) + \vartheta_a, \quad (87)$$

where $\vartheta_a = \sum_{i=1}^N \vartheta_{ia}$, and $\vartheta_{ia} \geq \sum_{i=1}^N \|W_{i\psi}^* S_{i\psi}\|^2 + 1/2 \sum_{i=1}^N \|F_{ie\psi}\|^2 + 1/2 \sum_{i=1}^N \|\dot{u}_{i0\psi}\|^2 + \gamma_{1ai}$. The definitions of symbols γ_{1ai} and ϵ_{3ai} refer to Remark 2.

Choose the following term:

$$\alpha_a = \sum_{i=1}^N \alpha_{ia} = \sum_{i=1}^N \min \{ 2(K_{ia} - 7/4), l_{ic\psi} \lambda_{\min}(S_{i\psi}), l_{ic\psi} \lambda_{\min}(S_{i\psi}), 2\epsilon_{3ai} \},$$

Table 1
Parameter values of UAVs ($i = 1, 2, 3, 4$).

Parameter	Value	Parameter	Value
m_i	2 kg	Q_{ix}	0.01 N s rad ⁻¹
l_i	0.2 m	Q_{iy}	0.01 N s rad ⁻¹
g	9.8 m/s ²	Q_{iz}	0.01 N s rad ⁻¹
I_{i1}	1.5 N s ² rad ⁻¹	$Q_{i\phi}$	0.015 N s rad ⁻¹
I_{i2}	1.5 N s ² rad ⁻¹	$Q_{i\theta}$	0.015 N s rad ⁻¹
I_{i3}	1.8 N s ² rad ⁻¹	$Q_{i\psi}$	0.015 N s rad ⁻¹

Table 2
Initial values of variables and leader information ($i = 1, 2, 3, 4$).

Variable	Value	Variable	Value
$q_1(0)$	[0.3 m, 0.3 m, 0.3 m] ^T	$\mathbf{q}_1(0)$	[$\frac{\pi}{12}$ rad, $\frac{\pi}{12}$ rad, $\frac{\pi}{2}$ rad] ^T
$q_2(0)$	[0.8 m, 0.3 m, 0.3 m] ^T	$\mathbf{q}_2(0)$	[$\frac{\pi}{12}$ rad, $\frac{\pi}{12}$ rad, $\frac{\pi}{2}$ rad] ^T
$q_3(0)$	[1.3 m, 0.3 m, 0.3 m] ^T	$\mathbf{q}_3(0)$	[$\frac{\pi}{12}$ rad, $\frac{\pi}{12}$ rad, $\frac{\pi}{2}$ rad] ^T
$q_4(0)$	[1.8 m, 0.3 m, 0.3 m] ^T	$\mathbf{q}_4(0)$	[$\frac{\pi}{12}$ rad, $\frac{\pi}{12}$ rad, $\frac{\pi}{2}$ rad] ^T
$v_1(0)$	[0.3 m/s, 0.3 m/s, 0.3 m/s] ^T	$w_1(0)$	[$\frac{\pi}{12}$ rad/s, $\frac{\pi}{12}$ rad/s, $\frac{\pi}{2}$ rad/s] ^T
$v_2(0)$	[0.3 m/s, 0.3 m/s, 0.3 m/s] ^T	$w_2(0)$	[$\frac{\pi}{12}$ rad/s, $\frac{\pi}{12}$ rad/s, $\frac{\pi}{2}$ rad/s] ^T
$v_3(0)$	[0.3 m/s, 0.3 m/s, 0.3 m/s] ^T	$w_3(0)$	[$\frac{\pi}{12}$ rad/s, $\frac{\pi}{12}$ rad/s, $\frac{\pi}{2}$ rad/s] ^T
$v_4(0)$	[0.3 m/s, 0.3 m/s, 0.3 m/s] ^T	$w_4(0)$	[$\frac{\pi}{12}$ rad/s, $\frac{\pi}{12}$ rad/s, $\frac{\pi}{2}$ rad/s] ^T
q_{i0}	[0.5 sin(t), 0.5 cos(t), 0.5] ^T	ψ_{i0}	0

then we get:

$$\dot{V}_\psi^\sigma \leq -\alpha_a V_\psi^\sigma + \vartheta_a. \quad (88)$$

Subsequently, by integrating Eq. (88) over time, it can be concluded that:

$$V_\psi^\sigma(t) \leq \exp(-\alpha_a t) V_\psi^\sigma(0) + \frac{\vartheta_a}{\alpha_a} (1 - \exp(-\alpha_a t)). \quad (89)$$

Define $L^{\sigma-1} = h_1^\sigma$, and referring to the existing results (57)–(61), the following conclusions can be easily drawn:

$$\begin{cases} \|Y_{\psi t}^\sigma\|^2 \leq \exp(-\lambda_1(t-T)) \|Y_{\psi 0}^\sigma\|^2, \quad v_1 = 1, \quad \kappa_1 = 0, \\ \|Y_{\psi t}^\sigma\|^2 \leq \exp(\kappa_1) \exp(\kappa_1 \mathbf{M}_0) \exp((\kappa_1/\eta_{\kappa_1} - \lambda_1)(t-T)) \|Y_{\psi 0}^\sigma\|^2, \quad \kappa_1 > 0, \quad \eta_{\kappa_1} > \kappa_1/\lambda_1. \end{cases} \quad (90)$$

According to Eqs. (89) and (90), we know that the states e_{iq} and $e_{i\omega}$ can converge to a bounded value close to the origin, as well as any error signal can be SGUUB, that ends the proof.

Based on the proofs of Theorems 1 and 2, it can be concluded that the proposed RL-based control method can achieve optimal robust formation tracking control for the given quadrotor multi-UAV systems.

Remark 5. According to Remark 1, it can be inferred that the given RL-based algorithm ((38)–(41) and (74)–(77)) can not only solve the optimal solution difficulties caused by complex nonlinearity but also, compared with other RL-based methods, release PE constraint.

Remark 6. This article describes the construction of the UDE observer using the filter G_{fi} . It is then integrated into the actor network to address model uncertainty and external disturbances. This method does not require parameter adjustment or prior information on controlled nominal models and external disturbances. Instead, it only needs appropriate bandwidth adjustments to the filter to eliminate complex unknown disturbances. This makes the controller design more straightforward.

5. Simulations

5.1. Verification results of the proposed controller

In this section, the MATLAB simulations of four quadcopter systems ($N = 4$) are presented, with 30 s, to verify the effectiveness of the proposed control algorithm.

The specific parameter values of the i th quadrotor multi-UAV systems (14) are selected as shown in Table 1 ($i = 1, 2, 3, 4$).

The initial values of variables and leader information are shown in Table 2.

Then, the formation intervals are selected as: $h_1 = [1, 1, 1]^T$, $h_2 = [2, 2, 2]^T$, $h_3 = [3, 3, 3]^T$, and $h_4 = [4, 4, 4]^T$.

The four drones are subjected to different interferences, and the corresponding simulation values are shown in Table 3.

Select parameters: $k_{ip} = 1$, $k_{ia} = 5$, and $K_{ip} = 2$, and $K_{ia} = 10$, and the initial values of critic-actor weights are chosen as $\hat{W}_{icp}(0) = [0.5]_{8 \times 3}$, $\hat{W}_{iap}(0) = [0.5]_{8 \times 3}$, $\hat{W}_{ic\psi}(0) = [0.5]_{8 \times 1}$, and $\hat{W}_{iap}(0) = [0.5]_{8 \times 1}$. The parameters of the neural network are given as $l_{icp} = 6$, $l_{iap} = 8$, $l_{ic\psi} = 6$, and $l_{i\psi} = 8$. The filter is designed in the following form:

$$G_{fi}(s) = \frac{1}{\rho s + 1}, \quad \rho = 0.001.$$

Table 3
Model uncertainties and external disturbances ($i = 1, 2, 3, 4$).

Variable	Value	Variable	Value
Δ_{ix}	$0.1x_i^2$	$\Delta_{i\phi}$	$0.1\phi_i + 0.1\dot{\phi}_i$
Δ_{iy}	$0.1y_i^2$	$\Delta_{i\theta}$	$0.1\theta_i + 0.1\dot{\theta}_i$
Δ_{iz}	$0.1z_i^2$	$\Delta_{i\psi}$	$0.1\psi_i + 0.1\dot{\psi}_i$
d_{1x}	$5 \sin(2t) + 5 \cos(t)$	$d_{1\phi}$	$5 \sin(2t) + 5 \cos(t)$
d_{1y}	$7.5 \sin(t) \cos(t)$	$d_{1\theta}$	$7.5 \sin(t) \cos(t)$
d_{1z}	$5 \sin(0.5t) + 5 \cos(0.5t) + 5$	$d_{1\psi}$	$5 \sin(0.5t) + 5 \cos(0.5t) + 5$
d_{2x}	$5 \sin(t)$	$d_{2\phi}$	$5 \sin(t)$
d_{2y}	$5 \cos(t)$	$d_{2\theta}$	$5 \cos(t)$
d_{2z}	$5 \sin(t) \cos(t) + 5 \cos(t) + 10$	$d_{2\psi}$	$5 \sin(t) \cos(t) + 5 \cos(t) + 10$
d_{3x}	$5 \sin(0.5t) \cos(0.5t)$	$d_{3\phi}$	$5 \sin(0.5t) \cos(0.5t)$
d_{3y}	$7.5 \sin(0.5t) \cos(0.5t)$	$d_{3\theta}$	$7.5 \sin(0.5t) \cos(0.5t)$
d_{3z}	$10 \sin(0.5t) \cos(0.5t)$	$d_{3\psi}$	$10 \sin(0.5t) \cos(0.5t)$
d_{4x}	$7.5 \sin(t) + 5$	$d_{4\phi}$	$7.5 \sin(t) + 5$
d_{4y}	$7.5 \cos(t) + 5$	$d_{4\theta}$	$7.5 \cos(t) + 5$
d_{4z}	$7.5 \sin(t) \cos(t) + 5$	$d_{4\psi}$	$7.5 \sin(t) \cos(t) + 5$

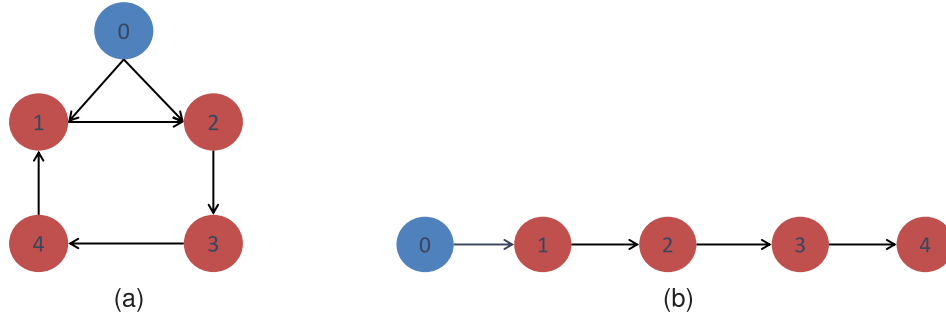


Fig. 3. Switching communication topologies.

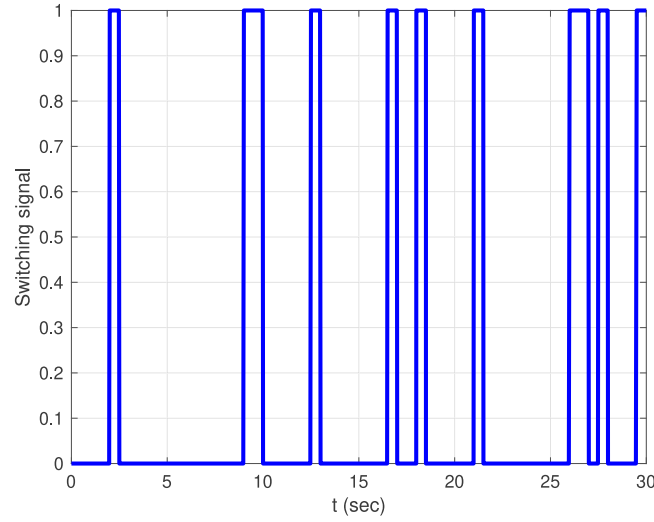


Fig. 4. Switching signal.

There are two directed communication topologies between drones that constantly switch over time, as shown in Fig. 3, with switching signal $\sigma = 1, 2$, which is described in Fig. 4, and the specific switching parameters are selected as 0.5 s sampling time, 0.3 maximum limits of the random number, and 0.25 s minimum dwell time. The corresponding Laplace matrix relationship is written as follows:

$$\begin{bmatrix} 0 & -1 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 0 \\ 0 & 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & -1 & 0 & 0 & 1 \end{bmatrix} \leftrightarrow \begin{bmatrix} 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

Other simulation diagrams are listed below.

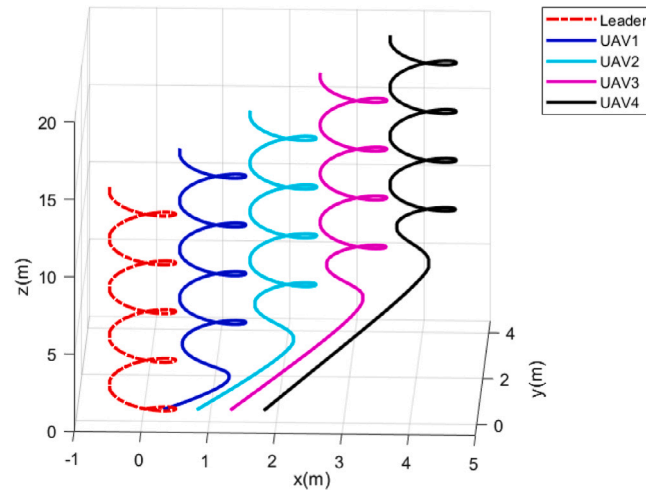


Fig. 5. Three-dimensional formation trajectories.

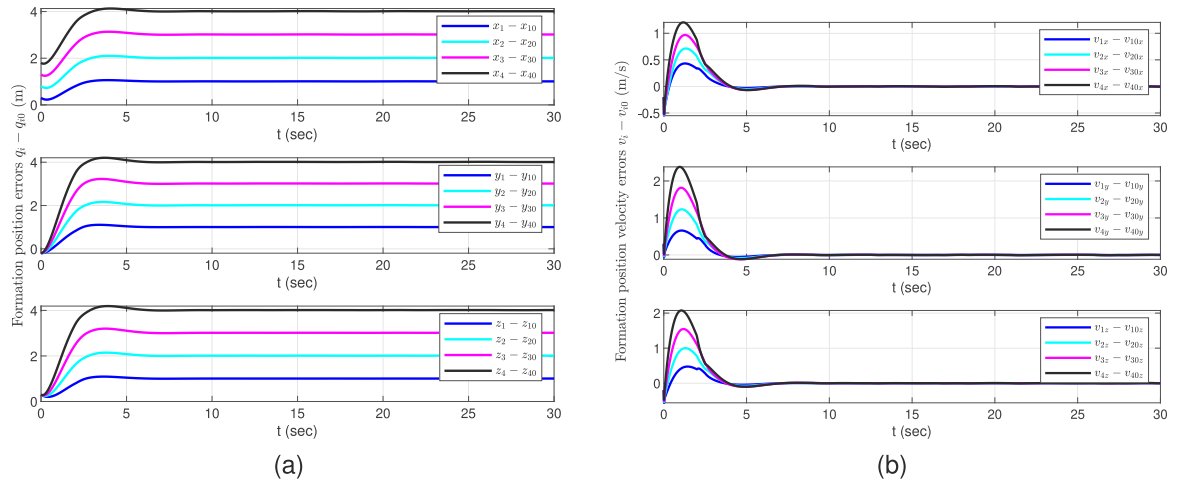


Fig. 6. Position tracking errors for four UAVs.

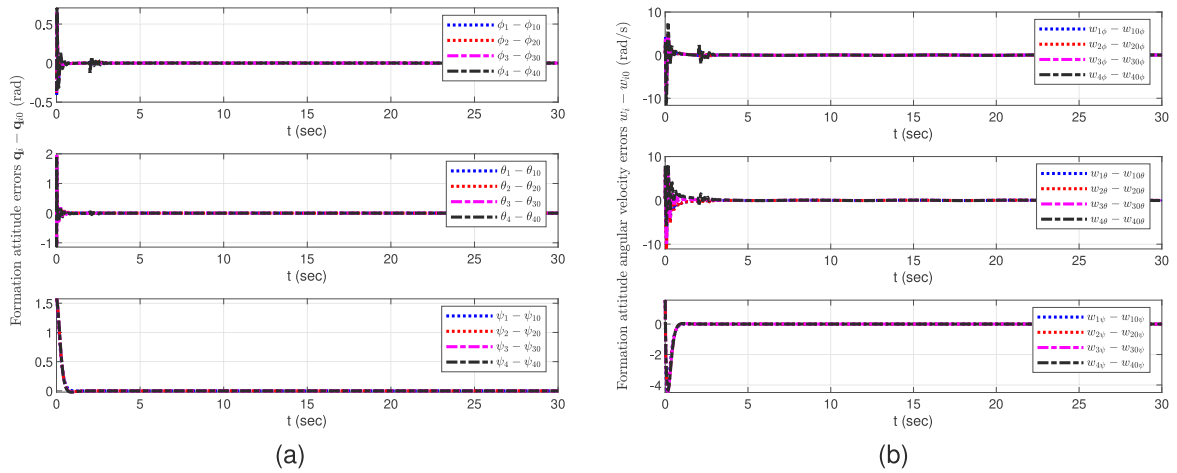


Fig. 7. Attitude tracking errors for four UAVs.

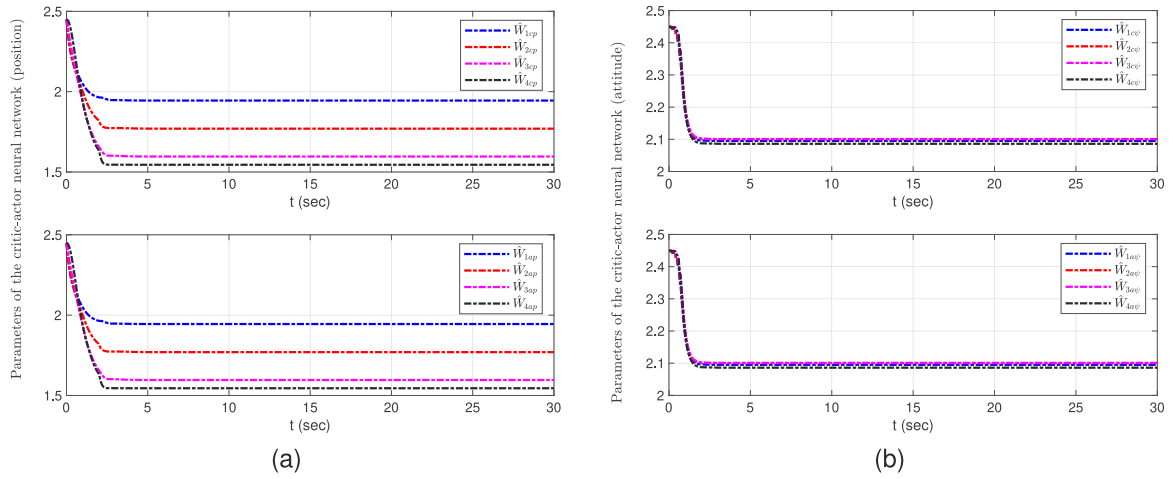


Fig. 8. Weights of the critic and the actor networks.

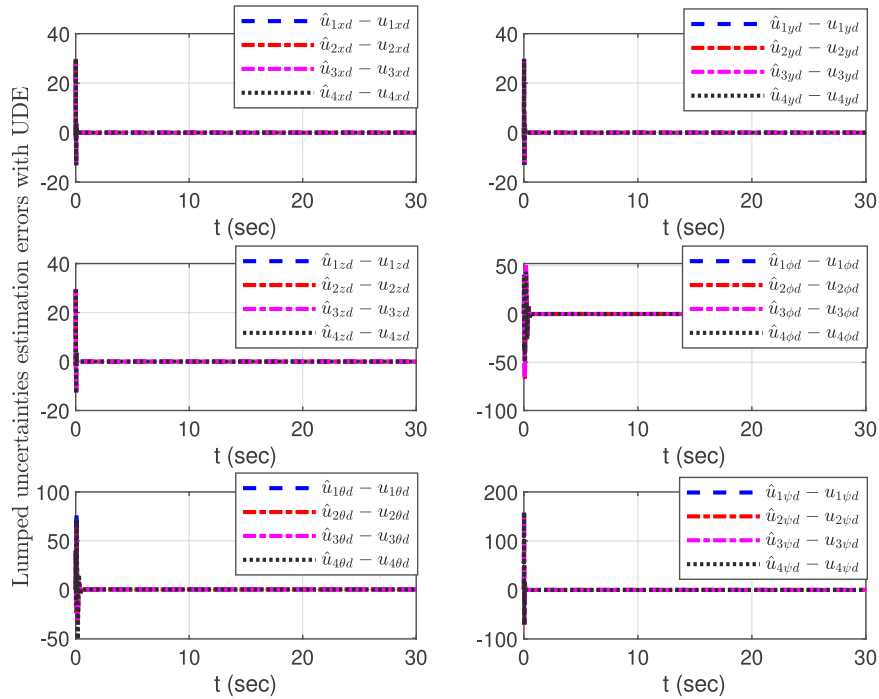


Fig. 9. Tracking errors of UDE observers for lumped uncertainty.

Figs. 5, 6, and 7 show that each drone has achieved the desired formation flight target. Fig. 8 illustrates that the weights of the RL neural network are bounded. Fig. 9 verifies that UDE has good robustness to model uncertainty and external disturbances on controlled systems. The above fully validates the effectiveness of the proposed RL-based optimized formation tracking control method.

5.2. Comparisons

As Remark 5 has fully demonstrated the advantages of the designed optimization algorithm, we present a series of comparative simulation experiments to illustrate the strong robustness of the given UDE observer under consistent switching communication topology and neural network training methods.

The control group consists of an RBF neural network method [16] and an adaptive compensator method [17], respectively. The compared state vector is the yaw angle ψ_i of the quadcopter. All parameter information refers to Section 5.1.

- (1). The controller with the adaptive compensator is designed as follows:

$$\tau_{ic\psi} = -K_{ia} Y_{i\psi}^\sigma - \frac{1}{2} \hat{W}_{ia\psi}^T S_{i\psi} + u_i, \quad i = 1, 2, 3, 4, \quad \sigma = 1, 2,$$

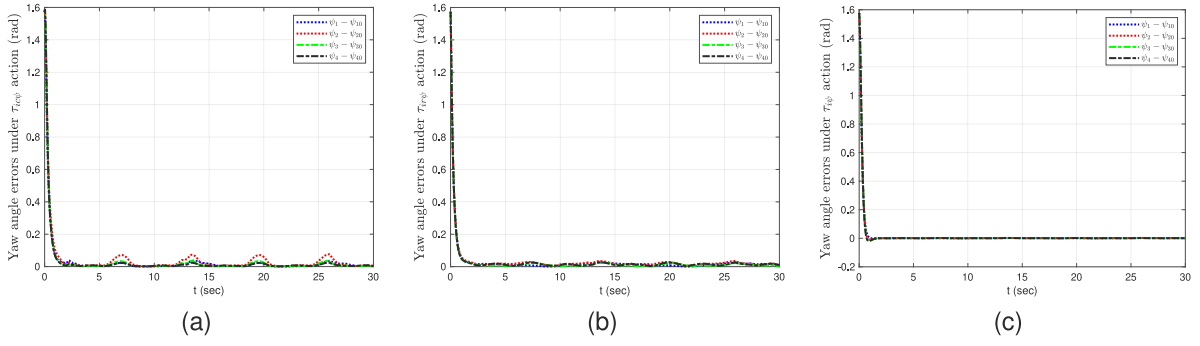


Fig. 10. (a) Yaw-angle errors of four drones under $\tau_{ic\psi}$ control. (b) Yaw-angle errors of four drones under $\tau_{ir\psi}$ control. (c) Yaw-angle errors of four drones under proposed $\tau_{i\psi}$ control.

where $u_i \in R$ is shown as:

$$u_i = -\frac{Y_{i\psi}^\sigma \Theta_i^2}{\|Y_{i\psi}^\sigma\| \Theta_i + \aleph_i},$$

and the update rate is designed as follows:

$$\dot{\aleph}_i = -l_i \aleph_i,$$

where $l_i = 10$, $\aleph_i(0) = 0.5$, and Θ_i is an estimation of lumped boundary value, which is provided as follows:

$$\dot{\Theta}_i = -k_d \|Y_{i\psi}^\sigma\|,$$

where $k_d = 1.5$.

(2). The controller with RBF neural network is built as follows:

$$\tau_{ir\psi} = -K_{ia} Y_{i\psi}^\sigma - \frac{1}{2} \hat{W}_{ia\psi}^T S_{i\psi} + \hat{W}_{ir\psi}^T \bar{S}_{i\psi}, \quad i = 1, 2, 3, 4, \quad \sigma = 1, 2,$$

where $\bar{S}_{i\psi}$ represents the Gaussian radial basis function expressed as $\bar{S}_{i\psi} = \exp\left(-\frac{\|Y_{i\psi}^\sigma - \bar{c}_J\|^2}{2D_J^2}\right)$, and the weight rate with $\mathbb{K}_i$ is chosen as:

$$\dot{\hat{W}}_{ir\psi} = \mathbb{K}_i Y_{i\psi}^{\sigma T} \bar{S}_{i\psi},$$

where the parameters are set as $\hat{W}_{ir\psi}(0) = [0.5]_{8 \times 3}$, $\mathbb{K}_i = 10$, and $D_J = 10$. $\bar{c}_J$ is designed as a random value, in which $J = 1, 2, \dots, 8$.

Afterwards, the results are shown in the following figures.

By observing Fig. 10, it is evident that the proposed control algorithm has better stability compared to the other two methods.

6. Conclusions

A novel RL-based formation controller for quadrotor multi-UAV systems has been proposed in this paper. This optimal control strategy designs a critic-actor neural network with a specific gradient descent training algorithm that can release the PE condition. It integrates the UDE observer method, which can effectively handle complex model uncertainty and unknown external disturbances. In addition, an adaptive sliding mode tracker based on the ADT technique is designed to switch the information exchange topology between drones and achieve steady-state control. The effectiveness and feasibility of the proposed control method have been validated using Lyapunov analysis and simulation examples. In the future, this work will expand to field experiments and develop its potential for engineering applications.

CRediT authorship contribution statement

Hongsheng Sha: Writing – original draft, Software, Methodology. **Rongwei Guo:** Visualization, Investigation. **Jin Zhou:** Conceptualization. **Xiaojin Zhu:** Validation. **Jinchen Ji:** Supervision. **Zhonghua Miao:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Hongsheng Sha received an M.S. degree from the Qilu University of Technology. He is pursuing a doctoral degree in engineering at Shanghai University. His research interests include multi-agent system consensus control and robust control.



Rongwei Guo received the B.S. degree from the University of Jinan, China, in 2001, the M.S. degree from Shanghai University, China, in 2004, and the Ph.D. degree from Shandong University, China, in 2011. He is currently a Professor with the School of Mathematics and Statistics, Qilu University of Technology (Shandong Academy of Sciences). His research interests include nonlinear systems control and switched systems.



Jin Zhou received his Ph.D. degree in operations research and cybernetics from Shanghai University, China, in 2003. He is currently a Full Professor and Doctoral Advisor of Shanghai Institute of Applied Mathematics and Mechanics, School of Mechanics and Engineering Science at Shanghai University. His research interests include complex systems and complex networks, cooperative control in networked multi-agent systems, and differential equations and system controls.



XiaoJin Zhu received his Ph.D. degree in mechatronic engineering from Nanjing University of Aeronautics and Astronautics, China, in 1998. He is currently a Full Professor and Doctoral Advisor in the School of Mechatronic Engineering and Automation at Shanghai University. His research interests include advanced measurement and control and information processing, intelligent structure active monitoring, computer measurement and control system and integration.



Jinchen Ji received his Ph.D. degree in mechanical engineering and is currently an Associate Professor of mechanical engineering at University of Technology Sydney, Australia. His research interests include dynamics and control of complex nonlinear systems and multi-agent systems, nonlinear dynamics and control of mechanical systems, vehicle system dynamics, and wind turbine dynamics.



Zhonghua Miao received his Ph.D. degree in mechatronic engineering from Shanghai Jiao-Tong University, China, in 2010. He is currently a Full Professor and Doctoral Advisor in the School of Mechatronic Engineering and Automation at Shanghai University. His research interests include intelligent robotics, measurement and control, and fault diagnosis, and agricultural machinery equipment.